Original Instructions

OPERATING MANUAL

CV5-500

ROBOT INTERFACE

(ADVANCED, ADDRESS SELECTION
METHOD SUPPORTED)

Manual No.: UD34SB0012E

NC: MAZATROL SmoothX

Serial No.:

Before using this machine and equipment, fully understand the contents of this manual to ensure proper operation. Should any questions arise, please ask the nearest Technical Center or Technology Center.

IMPORTANT NOTICE

1. **Be sure to observe the safety precautions described in this manual and the contents of the safety plates on the machine and equipment. Failure may cause serious personal injury or material damage. Please replace any missing safety plates as soon as possible.**
2. **No modifications are to be performed that will affect operation safety.**
3. **For the purpose of explaining the operation of the machine and equipment, some illustrations may not include safety features such as covers, doors, etc. Before operation, make sure all such items are in place.**
4. **This manual was considered complete and accurate at the time of publication, however, due to our desire to constantly improve the quality and specification of all our products, it is subject to change or modification. If you have any questions, please contact the nearest Technical Center or Technology Center.**
5. **Always keep this manual near the machinery for immediate use.**
6. **If a new manual is required, please order from the nearest Technical Center or Technology Center with the manual No. or the machine name, serial No. and manual name.**
7. **The export of machining programs may be restricted to prevent them from being used for military purposes. The machining programs you prepared are your property, so you should manage them yourself in accordance with the law.**

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12.2023

CONTENTS

1 OVERVIEW

1-1 Outline

This manual covers the robot interface provided on Mazak's machine tools.

This interface enables connection of the machine tool to a robot available from other manufacturers.

1-2 Terms

This section defines the terms used in detailed explanations in later sections.

- Machine

  Refers to a Mazak machine tool.

- Robot

  Refers to a device that is connected to the machine and used for loading a piece of material to be machined, for unloading a machined part, or for other purposes.

- Workpiece

  Refers generically to an object that the robot handles, including material to be machined and machined parts.

- ON/OFF

  For each of interface signals, the signal is said to be ON when the command is specified or the satisfying status is established for the signal. The signal is said to be OFF when no command is specified or the satisfying status is not established for the signal.

  - In the ON status, a voltage of 24 VDC is applied on the signal wire.
  - In the OFF status, the voltage on the signal wire is 0 V.

1-3 Application

The following shows the applicable combination of a machine and a communication system.

Type-A	Single pallet specification machine + field network communication

Note 1: In field network communication, it is assumed that the system is configured with the machine as a slave (an adapter).

Note 2: With regard to CC-Link, one of the field network communication systems, use a master unit that supports version 2.00 of CC-Link.

2 SAFETY PRECAUTIONS

This manual is prepared only for the function or the device which is specified in the title of this manual. Therefore, you are required to carefully read and understand the operating manual of the machine, especially the section describing safety precautions, before using or operating the function or the device.

2-1 Rules

The meanings of our safety precautions to DANGER, WARNING, and CAUTION are as follows:

DANGER : Failure to follow these instructions could result in loss of life.

WARNING : Failure to observe these instructions could result in serious harm to a human life or body.

CAUTION : Failure to observe these instructions could result in minor injuries or serious machine damage.

2-2 Safety Precautions Relating to the Function/Device Described in This Manual

WARNING

- When operating a robot using this interface, be sure to arrange a safety fence around the operating range of the machine and robot and input the open/closed signal of the safety fence to the specified terminals.

While this safety fence is closed, the following operations that will be required for setting up a workpiece using the robot can be performed, even with the front door of the machine open.

- Through-spindle air
- Spindle orientation
- Hydraulic fixture clamp/unclamp
- Front door open/close

CAUTION

- Do not use the 24 VDC power supplied at the NC unit other than for drivers and receivers, such as a clutch operation circuit.
- Do not connect the power supplied at the NC unit and the power provided for the robot side in parallel.
- The System Integrator is responsible for Robot Control, Safety Fence, Fixtures, and Fixture Control (in manual and automatic mode).
- Fixture Control is not part of the Robot Interface.

3 FLOW OF OPERATIONS FOR SYNCHRONIZING THE MACHINE AND THE ROBOT

This chapter describes the flow of operations from the installation of the machine and the robot to the start of their synchronized operation.

3-1 Connection of the Machine and the Robot

Connect the machine and the robot.

- For details, refer to the following sections.
  - 4-2 “Connection between the Machine and Robot”
  - 4-3 “Cable Connection”
  - 8-1 “Connections”

In the case of field network communications, the alarm **1362 ROBOT COMMUNICATION ERROR** will be raised after the connection if the button on the machine operating panel is enabled.

The alarm 1362 can be cleared after communication between the machine and the robot has been established.

3-2 Network Configuration

The operation covered in this section is not required for CC-Link.

For EtherNet/IP and PROFIBUS-DP communications, configure the network.

- For details, refer to 3-3-1 “Configuration for PROFIBUS-DP communications” and 3-3-2 “Configuration for EtherNet/IP communications” in the separate manual “Field Network Function (Option).”

3-3 Parameter Setting 1 of the Machine (Field Network Communications)

Set parameters for the machine.

- For details, refer to 8-4 “Parameters”.

3-4 Parameter Setting 2 of the Machine (For Field Network Communications)

Set parameters related to field network communication.

- For details, refer to 8-7 "Parameters Related to Field Network Input/Output Addresses (R Registers)".

Note : To determine the parameters, the following information is required.
Ask the robot manufacturer's person in charge of system integrator.

< EtherNet/IP and PROFIBUS-DP >

- Total data size of systems with a lower system number than the machine

< CC-Link >

- Station number assigned to the machine
- Transmission rate
- Number of exclusive stations in each system and extended cyclic settings
- Presence or absence of remote I/O stations in systems with a lower station number than the machine

3-5 Confirmation of the Establishment of Communication

If the button on the machine operating panel is enabled, the alarm **1362 ROBOT COMMUNICATION ERROR** can be cleared after communication between the machine and the robot has been established correctly.

If the alarm 1362 cannot be cleared, check:

1. Whether the robot has received the output signal DO 000 (communication check OUTPUT) from the machine.
   - If the robot has not received the signal, check Sections 3-2 through 3-4 again.
2. Whether the robot has output the input signal DI 000 (communication check INPUT) to the machine.
   - If the robot has not output the signal, modify the sequence program on the robot side.
3. Whether the machine has received the input signal DI 000 (communication check INPUT)
   - If the machine has not received the signal, check Sections 3-2 through 3-4 again.

3-6 Creation of a Machining Program

Create a machining program for the synchronized operation of the machine and the robot.

- For details, refer to 8-2 "Automatic Operation".

3-7 Start of the Synchronized Operation

When the button on the machine operating panel is enabled, the machine and the robot can be operated in synchronization.

4 INTERFACE CONNECTION

4-1 Communication System

The communication system applicable to the machine is field network communication.

Note : Check the type of communication system incorporated in the machine beforehand.

1. Field network communication

The following three types of field network are supported.

- PROFIBUS-DP (Mazak: Master)
- EtherNet/IP (Mazak: Scanner)
- CC-Link (Mazak: Master/Slave)

The following shows the range of R registers available for each field network protocol.

Refer to the addresses for CH1 when using expansion slot EXT3, or for CH2 when using expansion slot EXT4.

- For details on the port for expansion slots, refer to 8-1-2 "Field network cable".

A. Range of R registers available for PROFIBUS-DP and EtherNet/IP

1. Data output from the NC unit

Within the following range, a total of 100 bytes are occupied.

Address		Meaning		Description
CH1	CH2			
R18176 to R18815	R26876 to R27515	Peripheral device interface		Represents data for peripheral devices. **Note :** Details may vary depending on the machine model and system configuration. Be sure to check the specifications at the machine side.
		Unit	—	

2. Data input to the NC unit

Within the following range, a total of 100 bytes are occupied.

Address		Meaning		Description
CH1	CH2			
R16384 to R17023	R25084 to R25723	Peripheral device interface		Represents data for peripheral devices. **Note :** Details may vary depending on the machine model and system configuration. Be sure to check the specifications at the machine side.
		Unit	—	

The area of registers to be used is determined by specifying the initial address with a parameter at the machine side. Set the machine side parameter so that the area is within the range given above.

- For details on the machine side parameter, refer to 8-4 “Parameters”.

B. Range of R registers available for CC-Link

Details of the R registers used for CC-Link are given below.

1. Data output from the NC unit

   Within the following range, 8 words (16 bytes) are occupied for remote output (Ry) and 16 words (32 bytes) for remote registers (RWw). (Number of occupied stations: 4, extended cyclic setting: single)

   The area occupied for remote output includes a 1-word (2 bytes) area that is occupied by the system when the local station is used.

<table>
<tr><th colspan="2">Address</th><th colspan="2" rowspan="2">Meaning</th><th rowspan="2">Description</th></tr>
<tr><th>CH1</th><th>CH2</th></tr>
<tr><td rowspan="2">R18176
to
R18527
R18816
to
R19455</td><td rowspan="2">R26876
to
R27227
R27516
to
R27155</td><td colspan="2">Peripheral device interface</td><td rowspan="2">Represents data for peripheral devices.

Note :
Details may vary depending on the machine model and system configuration. Be sure to check the specifications at the machine side.</td></tr>
<tr><td>Unit</td><td>—</td></tr>
</table>

2. Data input to the NC unit

   Within the following range, 8 words (16 bytes) are occupied for remote input (Rx) and 16 words (32 bytes) for remote registers (RWr).

   The area occupied for remote input includes a 1-word (2 bytes) area that is occupied by the system when the local station is used.

<table>
<tr><th colspan="2">Address</th><th colspan="2" rowspan="2">Meaning</th><th rowspan="2">Description</th></tr>
<tr><th>CH1</th><th>CH2</th></tr>
<tr><td rowspan="2">R16384
to
R17663</td><td rowspan="2">R25084
to
R26363</td><td colspan="2">Peripheral device interface</td><td rowspan="2">Represents data for peripheral devices.

Note :
Details may vary depending on the machine model and system configuration. Be sure to check the specifications at the machine side.</td></tr>
<tr><td>Unit</td><td>—</td></tr>
</table>

The area of registers to be used is determined by specifying the initial address with a parameter at the machine side. Set the machine side parameter so that the area is within the range given above.

- For details on the machine side parameter, refer to 8-4 “Parameters”.

4-2 Connection between the Machine and Robot

1. Field network communication

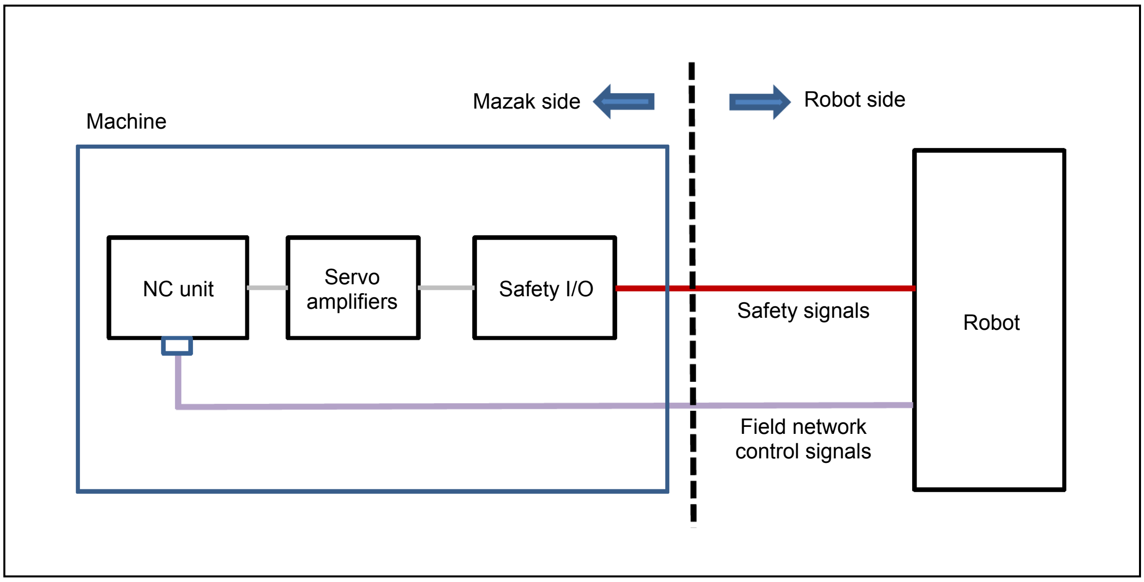


Fig. 4-1 Field network communication

4-3 Cable Connection

4-3-1 Field network communication

Cables used for field network communication are connected using terminal blocks and the connectors on the NC unit.

The cable for carrying field network communication signals is directly connected to the connector on the NC unit and the cables for carrying safety signals are connected through the terminal blocks.

Mazak takes responsibility for connection to the connectors on the machine and the terminal blocks, while the connection from the connectors and terminal blocks to the robot is the customer's responsibility.

Cables and connectors outside the range of the machine's control and the power supply to the robot need to be prepared by the customer.

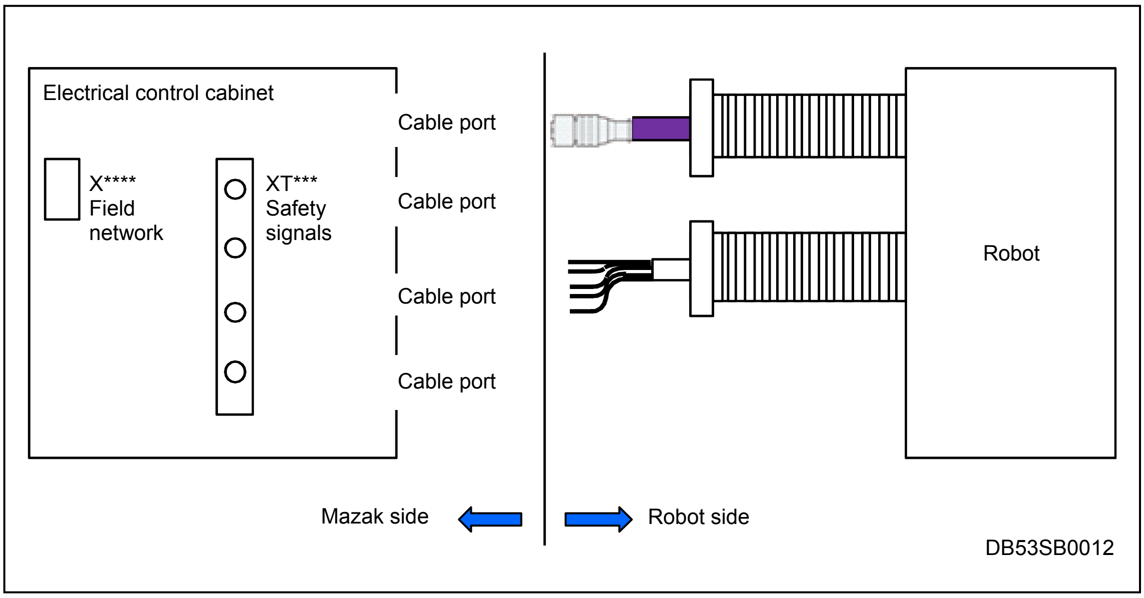


Fig. 4-2 Field network communication cable connection

	• The code indicated at "****" differs depending on the machine model. For the device number, refer to the electrical wiring diagrams of the machine used.

1. X****: Field network connectors

Connectors appropriate for each network device are used in field network communication.

A. CC-Link

1. Machine side connector

   Maker: 3M

   Model: 35610-5253-B00PE

   For details, refer to the manufacturer’s catalog.

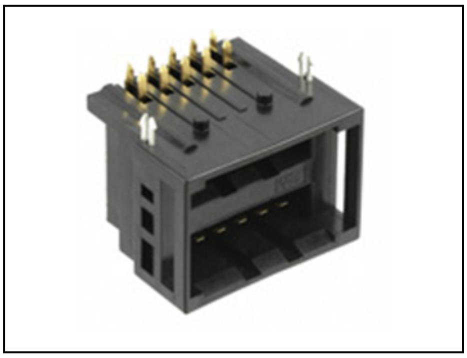


Fig. 4-3 CC-Link connector at the machine side

2. Cable side connector

   Maker: 3M

   Model: 35505-6000-BOM GF

   For details, refer to the manufacturer’s catalog.

Fig. 4-4 CC-Link connector at the cable side

B. EtherNet/IP

1. Machine side connector

   Maker: HIROSE ELECTRIC Co., Ltd.

   Model: TM11R-5M3-88-LP

   For details, refer to the manufacturer’s catalog.

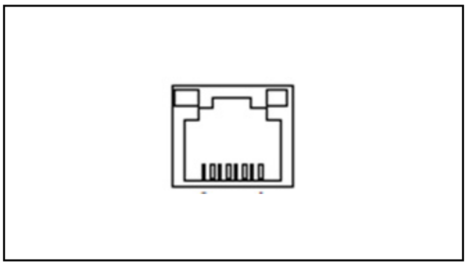


Fig. 4-5 EtherNet/IP connector at the machine side

2. Cable side connector

   Maker: Japan Telegärtner Ltd.

   Model: J00026A0165

   For details, refer to the manufacturer’s catalog.

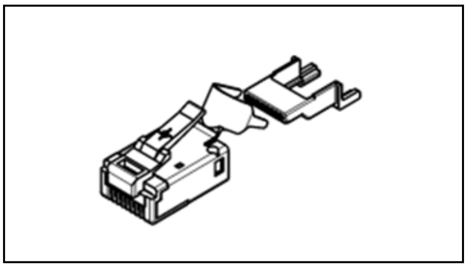


Fig. 4-6 EtherNet/IP connector at the cable side

C. PROFIBUS-DP

1. Machine side connector

   Maker: OMRON Corporation

   Model: XM3B-0922-132

   For details, refer to the manufacturer's catalog.

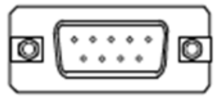


Fig. 4-7 PROFIBUS-DP connector at the machine side

2. Cable side connector

   Maker: Siemens AG

   Model: 6GK1 500-0FC10

   For details, refer to the manufacturer's catalog.

Fig. 4-8 PROFIBUS-DP connector at the cable side

2. Connector for Safety Signals

For the safety signal connection, an industrial rectangular connector is used.

1. Machine side connector

   Maker: ILME

   This connector is fitted to the electrical cabinet of the Mazak machine.

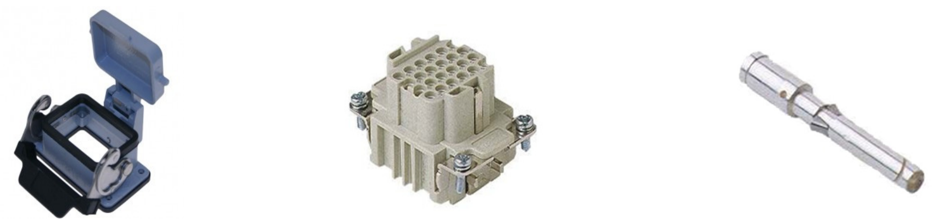


Fig. 4-9 Machine side connector

2. Cable side connector

   Maker: ILME

   This connector has to be provided by the Robot system.

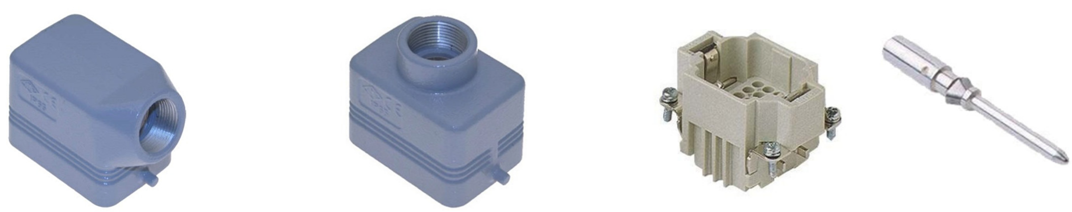


Fig. 4-10 Cable side connector

4-4 Signal Requirements

4-4-1 Safety signals

1. Signals input to the machine

With respect to safety, emergency stop signals and safety fence closed signals are critical. Configure an electrical circuit as shown below.

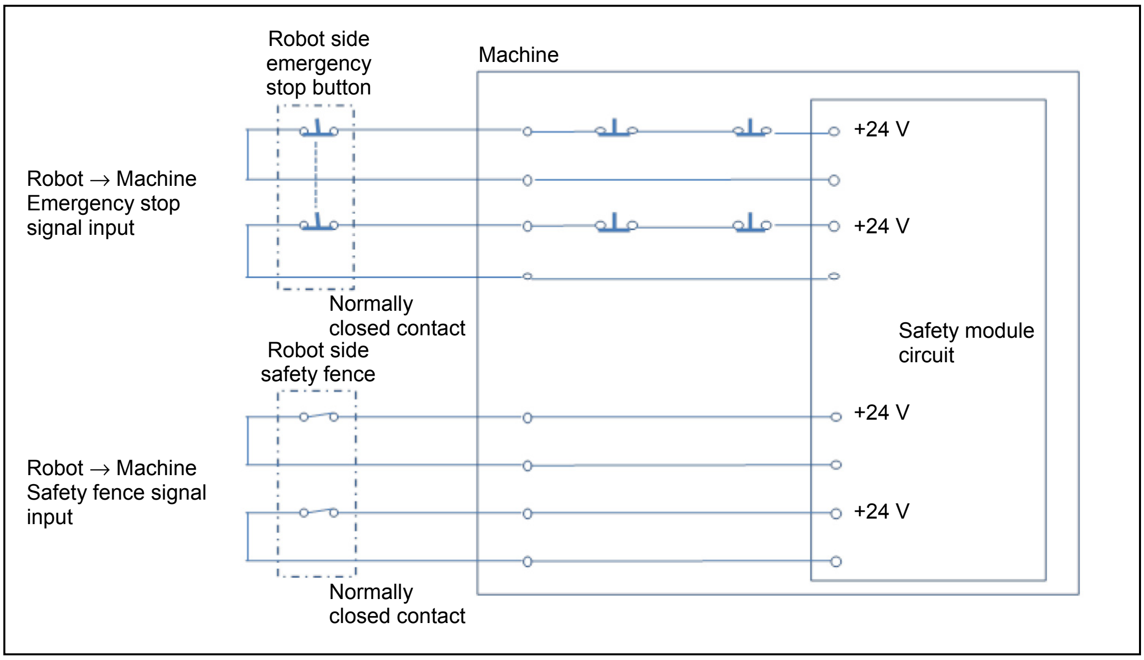


Fig. 4-11 Safety signals (input)

- Provide dry contacts (no-voltage contacts) for these input signals.
- These input signals need to be controlled using independent contacts.

2. Signals output from the machine

Emergency stop signals, front door closed and side door closed signals can be used as safety input signals for the robot.

Configure an electrical circuit as shown below.

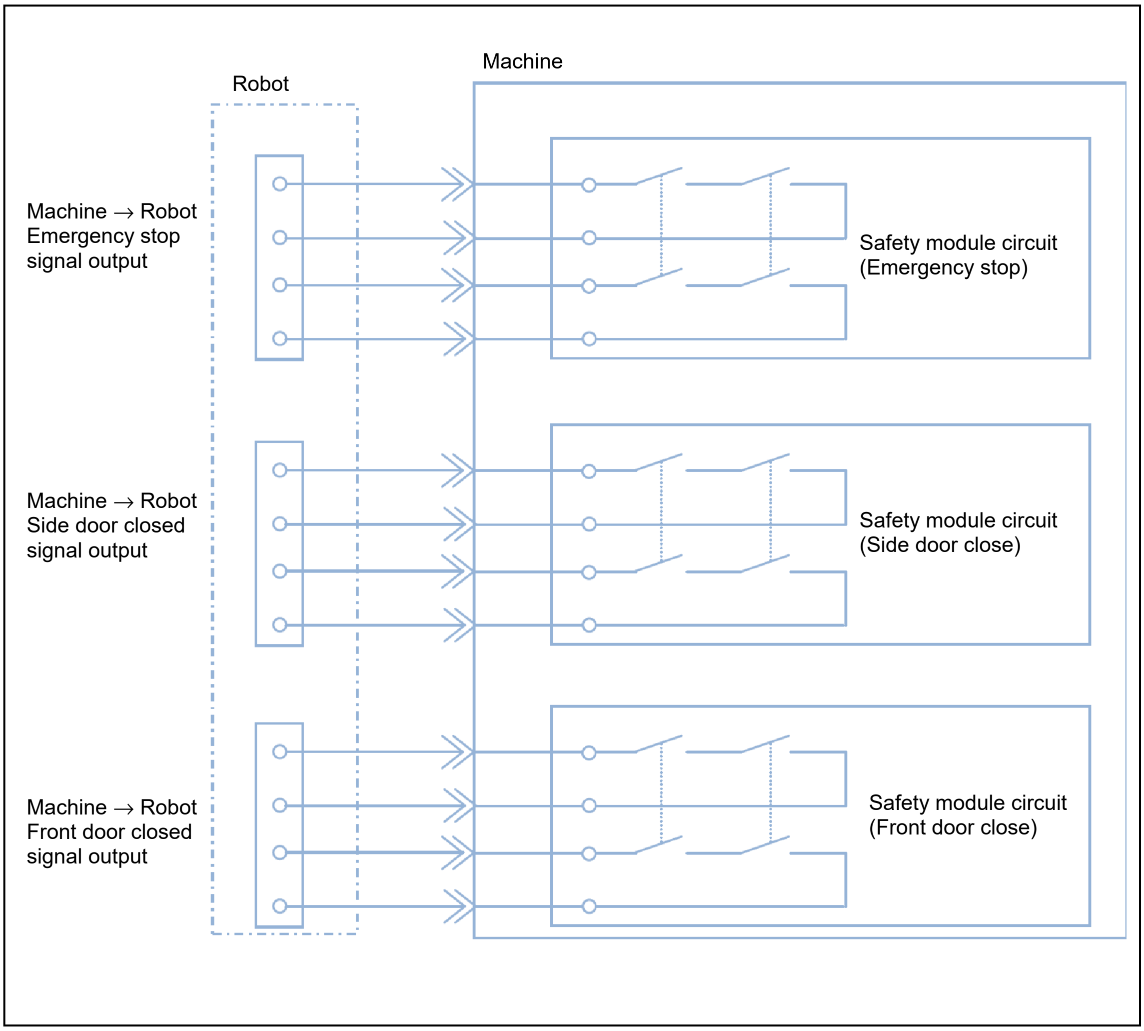


Fig. 4-12 Safety signals (output)

- Provide dry contacts (no-voltage contacts) for these output signals.
- These output signals are controlled using independent contacts.
- Rated load of the contacts: 250 VAC/6A, 30 VDC/6 A

3. Safety fence signals

Arrange safety fences that cover the operating range of the robot and the front door of the machine (or side door of the machine) as shown in the figure below and check that the safety fence door is closed using a switch or a similar device.

Note : Never use the machine with the system for checking the safety fence door status shortcircuited without installing a switch.

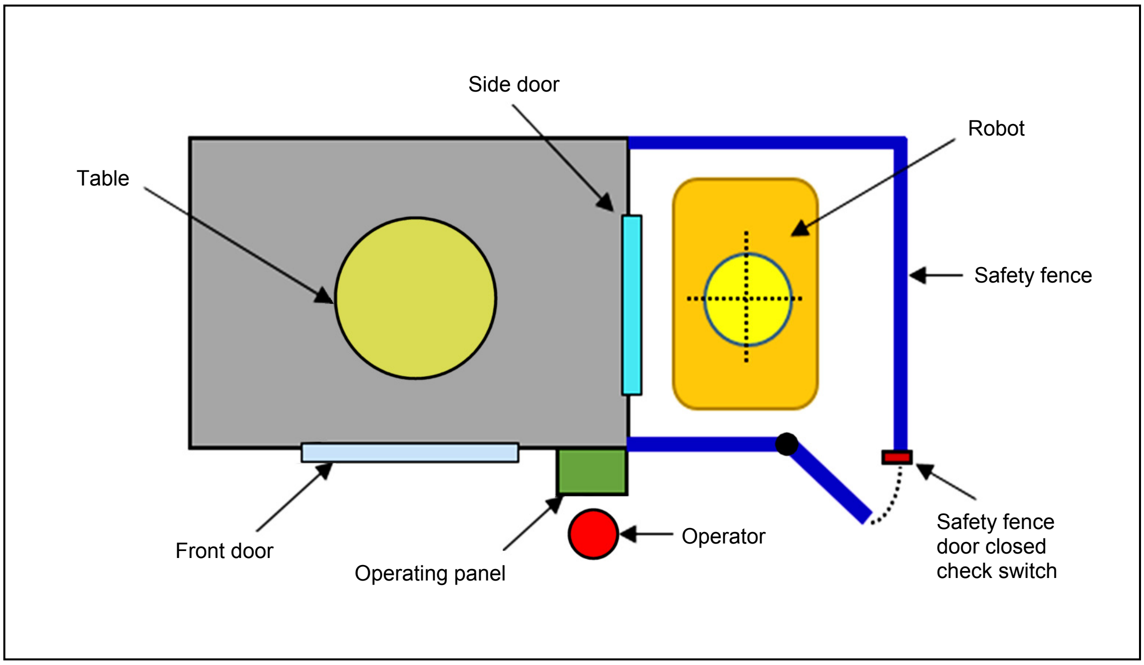


Fig. 4-13 Example installation of safety fences (using Side Door)

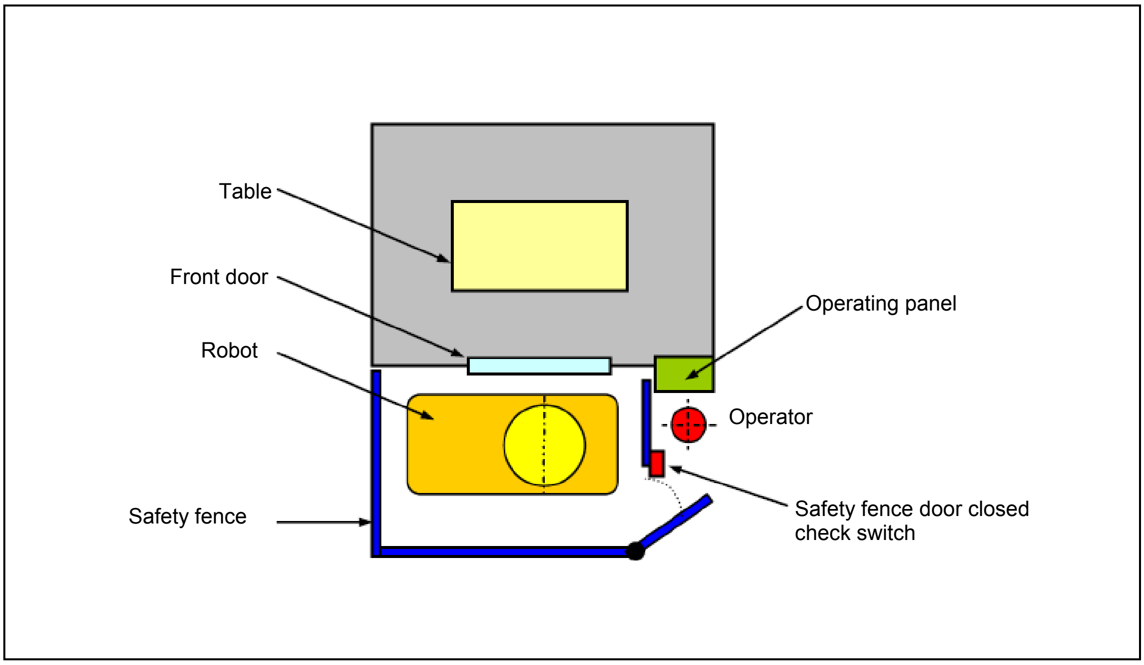


Fig. 4-14 Example installation of safety fences (using Front Door)

5 LIST OF SIGNALS

5-1 List of Input Signals (Robot → Mazak)

Machine Type

- A: Single pallet specification machine (field network)

Supported field network types

- PROFIBUS-DP
- Ethernet IP
- CC-Link

1. SI 000-SI 001: Safety signal input

●: Standard function ○: Optional function

No.	Signals	Availability	Remarks
		A	
SI 000	Emergency stop signal INPUT	●	**Note**
SI 001	Safety fence closed signal	●	**Note**

Note : These signals have to be provided as voltage free contacts.

- For details, refer to 4-4-1 “Safety signals”.

2. DI 000-DI 019: Standard interface input

●: Standard function ○: Optional function

No.	Signals	Data Size	Availability A	Remarks
DI 000	Communication check input	Bit	●	
DI 001	Robot Ready	Bit	●	
DI 002	Machine Stop Request	Bit	●	
DI 003	Robot Running	Bit	●	
DI 004	Robot Alarm State	Bit	●	
DI 005	Operator Intervention Request	Bit	●	
DI 006	Machine Power OFF request	Bit	●	
DI 007	(No allocation)	–	–	
DI 008	Fixture 1 Clamp Command	Bit	○	
DI 009	Fixture 1 Unclamp Command	Bit	○	
DI 010	Fixture 2 Clamp Command	Bit	○	
DI 011	Fixture 2 Unclamp Command	Bit	○	
DI 012 | DI 019	(No allocation)	–	–	

3. DI 100-DI 129: Standard interface input

●: Standard function ○: Optional function

No.	Signals	Data Size	Availability A	Remarks
DI 100	Target work number data	32 bytes	●	
DI 101	Work number search start	Bit	●	
DI 102	Cycle start command	Bit	●	
DI 103	NC reset	Bit	●	
DI 104	All machining complete	Bit	●	
DI 105	(No allocation)	–	–	
DI 106	Robot Service Finished Signal	Bit	●	
DI 107	Machine Front / Side door open command (**Note**)	Bit	●	
DI 108	Machine Front / Side door close command (**Note**)	Bit	●	
DI 109	Robot clear	Bit	●	
DI 110 | DI 129	(No allocation)	–	–	

Note : The machine can be loaded/ unloaded by the robot using the front door or the side door. When the front door is used, then the signals DI107 and DI108 refer to the front door. When the side door is used, then the signals DI107 and DI108 refer to the side door.

5-2 List of Output Signals (Mazak → Robot)

Machine Type

- A: Single pallet specification machine (field network)

Supported field network types

- PROFIBUS-DP
- Ethernet IP
- CC-Link

1. SO 000-SO 001: Safety signal output

●: Standard function ○: Optional function

No.	Signals	Availability A	Remarks
SO 000	Emergency stop signal OUTPUT	●	**Note**
SO 001	Side door closed	●	**Note**
SO 002	Machine front door closed	●	**Note**

Note : These signals have to be provided as voltage free contacts.

- For details, refer to 4-4-1 “Safety signals”.

2. DO 000-DO 019: Standard interface output

●: Standard function ○: Optional function

No.	Signals	Data Size	Availability A	Remarks
DO 000	Communication check output	Bit	●	
DO 001	Machine Ready	Bit	●	
DO 002	Robot Stop Request	Bit	●	
DO 003	Machine operating panel position	Bit	●	
DO 004	Machine Alarm State	Bit	●	
DO 005	(No allocation)	–	–	
DO 006	Automatic Power OFF acceptance	Bit	●	
DO 007	(No allocation)	–	–	
DO 008	Fixture 1 Clamped Finished Signal	Bit	○	
DO 009	Fixture 1 Unclamped Finished Signal	Bit	○	
DO 010	Fixture 2 Clamped Finished Signal	Bit	○	
DO 011	Fixture 2 Unclamped Finished Signal	Bit	○	
DO 012 | DO 019	(No allocation)	–	–	

3. DO 100-DO 129: Standard interface output

●: Standard function ○: Optional function

No.	Signals	Data Size	Availability A	Remarks
DO 100	Current Work Number	32 bytes	●	
DO 101	Work Number search finished signal	Bit	●	
DO 102	Cycle Start Enable	Bit	●	
DO 103	Machine Running	Bit	●	
DO 104	Machining complete	Bit	●	
DO 105	Robot service codes	4 bits	●	
DO 106	Robot service request	Bit	●	
DO 107	Machine Front / Side door open finished signal (**Note**)	Bit	●	
DO 108	Machine Front / Side door close finished signal (**Note**)	Bit	●	
DO 109	Robot access to machine permitted	Bit	●	
DO 110 | DO 129	(No allocation)	–	–	

Note : The machine can be loaded/ unloaded by the robot using the front door or the side door. When the front door is used, then the signals DO107 and DO108 refer to the front door. When the side door is used, then the signals DO107 and DO108 refer to the side door.

6 SIGNAL DETAILS

6-1 Details on Input Signals (Robot → Mazak)

1. SI 000: Emergency stop signal INPUT

A. Function

Used to stop the machine.

Note : This signal is used in a safety function of the machine. The stop category of this signal is category 3.

This signal is provided using a double dry contact (no-voltage contact).

- For details, refer to 4-4-1 “Safety signals”.

B. Input conditions and operation

- When this signal turns OFF (open), the machine stops in an emergency stop status.
- To enable machine operation, turn this signal ON (short).
- The function of this signal is equivalent to that of pressing the emergency stop button on the machine operating panel.

2. SI 001: Safety fence closed signal

A. Function

Turning this signal ON enables the following operations even with the machine’s front / side door not closed.

- Spindle rotation (spindle orientation/spindle air blast)
- Linear axis movement
- Front / Side door open/close operation

While this signal is OFF, the machine turns OFF the “DO 002: Robot stop request” signal to stop axis operation at the robot side.

Note : This signal is used in a safety function of the machine. The stop category of this signal is category 3.

This signal is provided using a double dry contact (no-voltage contact).

- For details, refer to 4-4-1 “Safety signals”.

B. Input conditions and operation

- When this signal is turned OFF while the front door / side door is not closed, the machine operation is stopped.

  At the same time, axis operation at the robot side is stopped using the “DO 002: Robot stop request” signal.
- When this signal is turned OFF while the front door / side door is not closed, the machine operation is not stopped.

  However, axis operation at the robot side is stopped using the “DO 002: Robot stop request” signal.

Turn this signal ON while the operator’s safety is secured by the safety fence or other safety devices.

< Safety fence closed signal (when the front door / side door is not closed) >

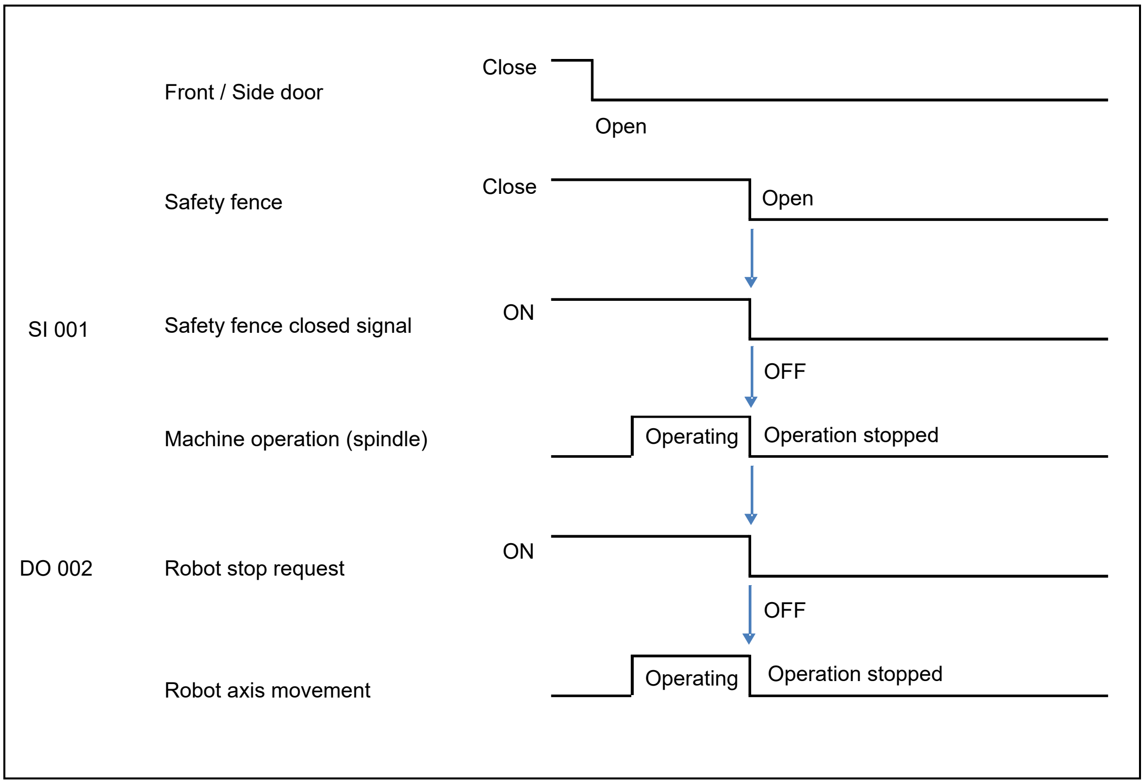


Fig. 6-1 Safety fence closed signal (when the front door / side door is not closed)

< Safety fence closed signal (when the front door / side door is closed) >

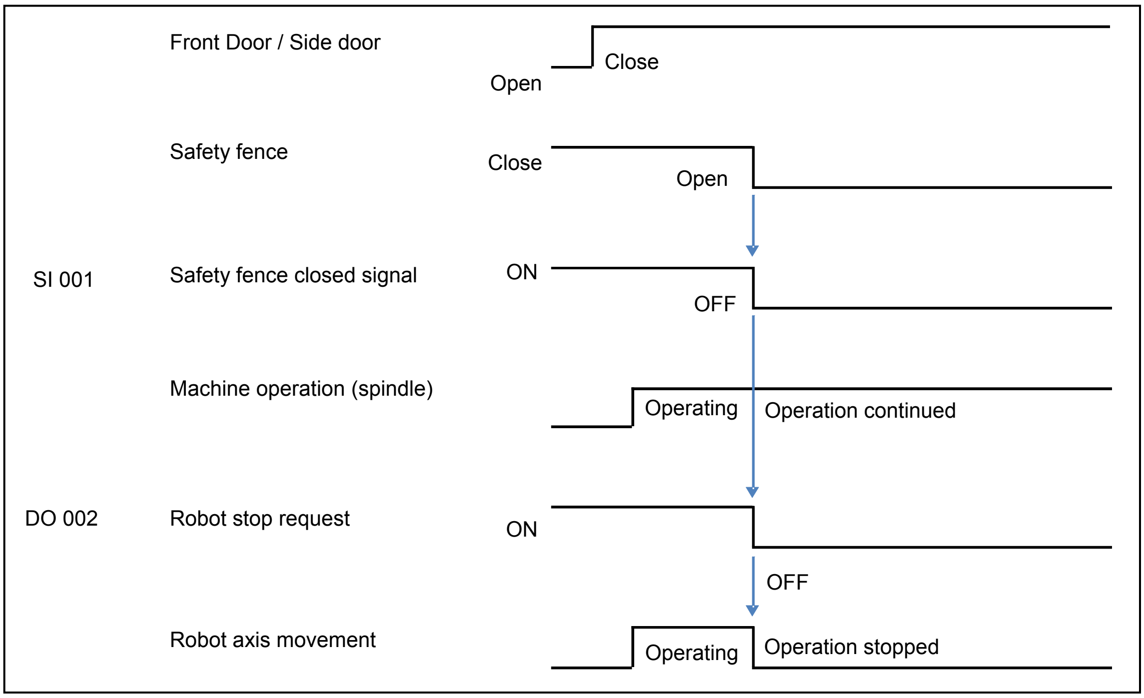


Fig. 6-2 Safety fence closed signal (when the front door / the side door is closed)

3. DI 000: Communication check INPUT

A. Function

The machine always monitors this signal to check whether the communication is carried out normally.

If switching of this signal does not occur for a certain period of time, a communication error between the robot and machine is assumed.

When this happens, the machine disables the robot interface.

B. Input conditions and operation

Turn this signal ON upon detecting turning ON of the “DO 000: Communication check OUTPUT” signal.

Turn this signal OFF upon detecting turning OFF of the “DO 000: Communication check OUTPUT” signal.

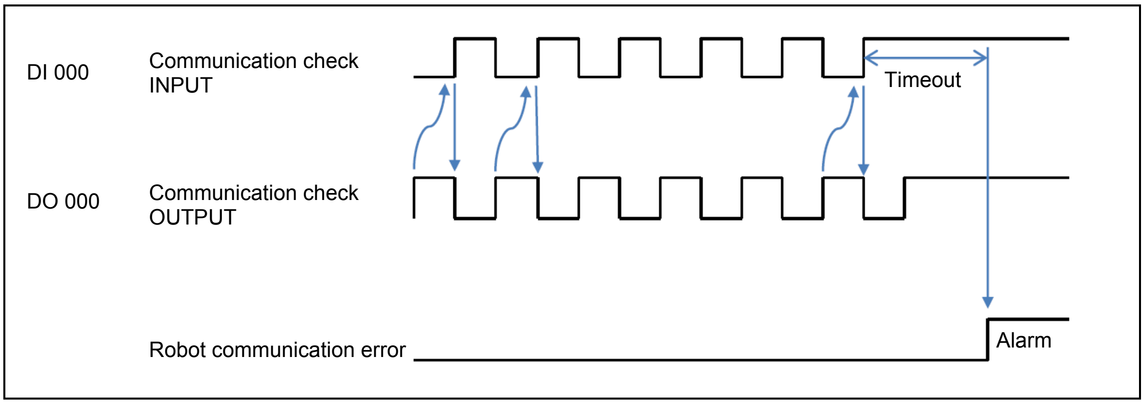


Fig. 6-3 Communication check INPUT signal

4. DI 001: Robot ready

A. Function

Turning this signal ON enables the robot interface function.

Turning this signal OFF enables independent operation of the machine.

B. Input conditions and operation

To use the robot interface function, this signal needs to be turned ON.

Turn this signal OFF when the robot is in an emergency stop status.

5. DI 002: Machine stop request

A. Function

Used to stop the machine in operation.

B. Input conditions and operation

Turning this signal OFF stops the following operations.

- Spindle rotation (spindle orientation/spindle air blast)
- Linear axis movement
- Front / Side door opening/closing operation

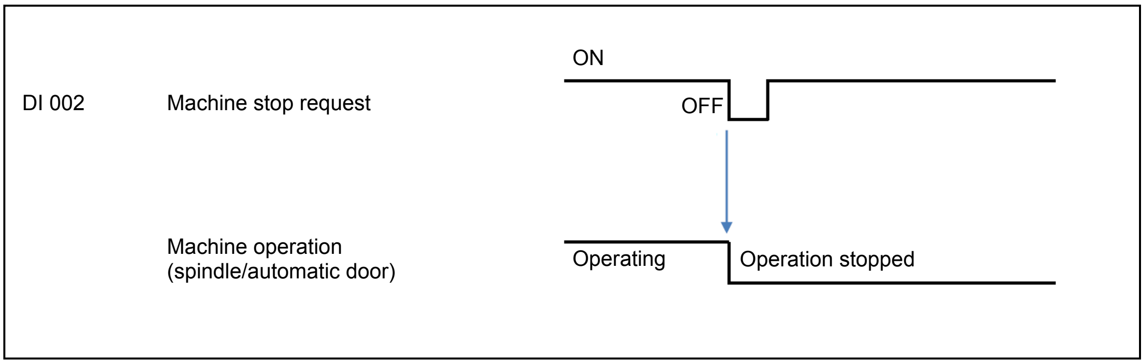


Fig. 6-4 Machine stop request signal

6. DI 003: Robot operating

A. Function

Used to notify the machine that axis operation of the robot is in progress.

B. How to use

This signal is used in response to the “DO 002: Robot stop request” signal.

C. Input conditions and operation

Turn this signal ON while axis operation of the robot is in progress.

Turn this signal OFF while the robot is stopped.

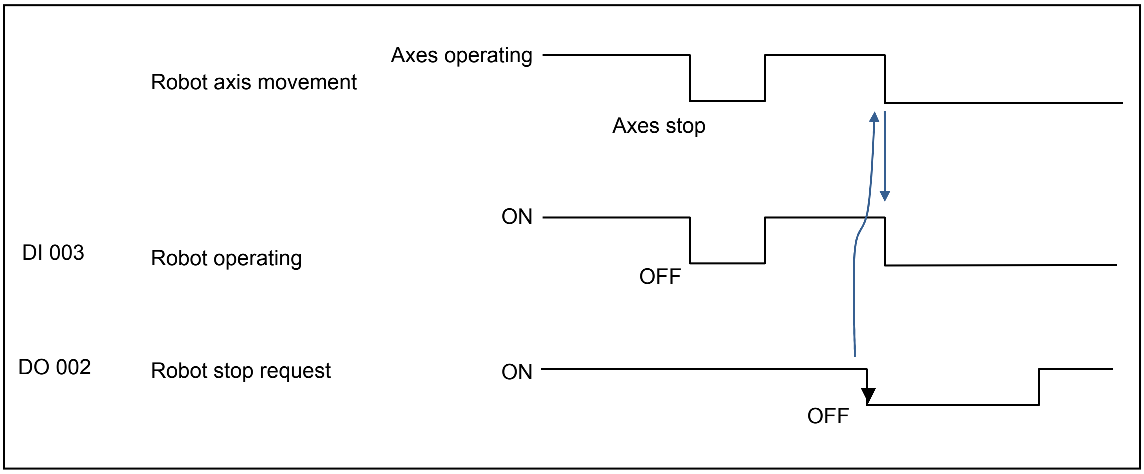


Fig. 6-5 Robot operating signal

7. DI 004: Robot alarm status

A. Function

Used to display a robot alarm on the machine screen.

B. Input conditions and operation

Turning this signal ON displays the alarm message on the machine screen.

It only displays the message and does not stop machine operation.

Turning this signal OFF turns off the message display.

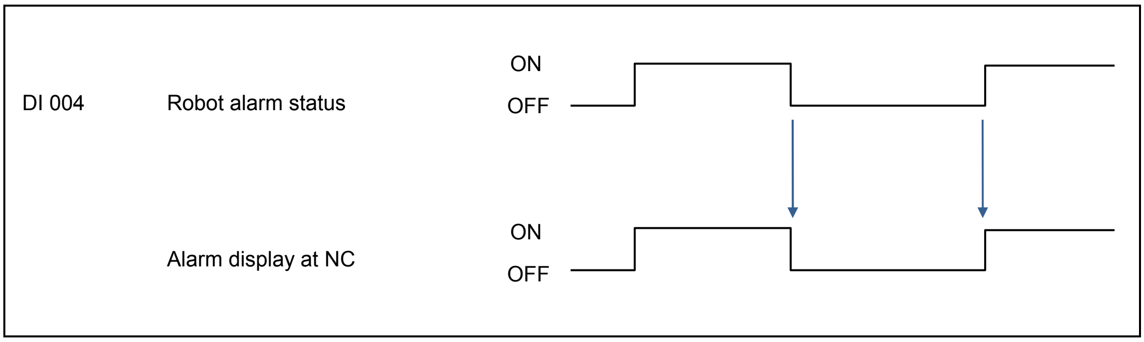


Fig. 6-6 Robot alarm status signal

8. DI 005: Operator interruption request

A. Function

Used to notify an operator interruption request.

B. Input conditions and operation

Turning this signal ON while the machine is waiting for the completion of a robot service M code extends the M code timeout period at the machine side.

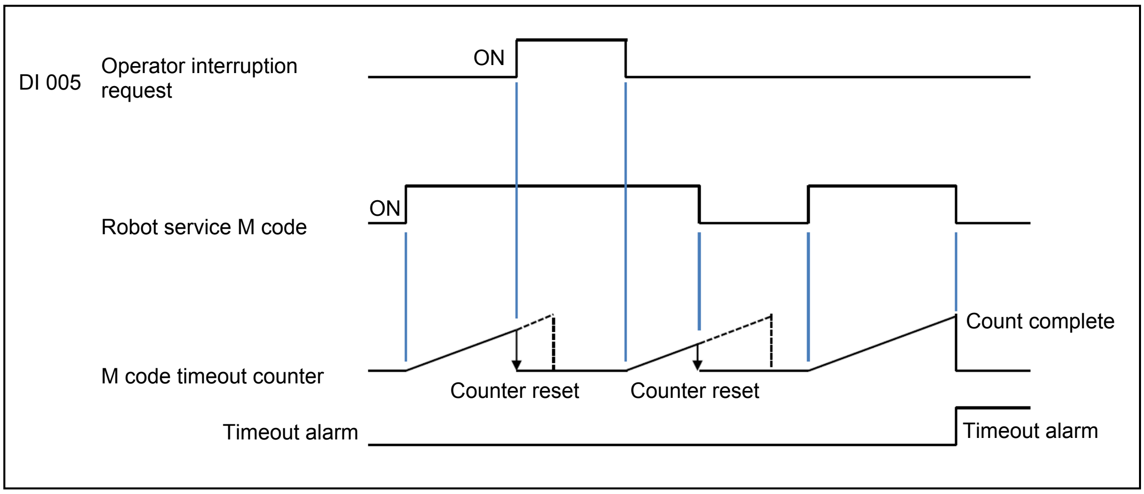


Fig. 6-7 Operator interruption request signal

9. DI 006: Automatic power shut-off request

A. Function

When the machine is equipped with the automatic power shut-off function, the machine power can be turned off from the robot side using this signal.

B. Input conditions and operation

Turning this signal ON sets a request to turn off the machine power.

The machine power is turned off if this signal is ON upon completion of the current machining process.

When turning the power off, the machine turns the “DO 006: Automatic power shut-off request received” signal ON.

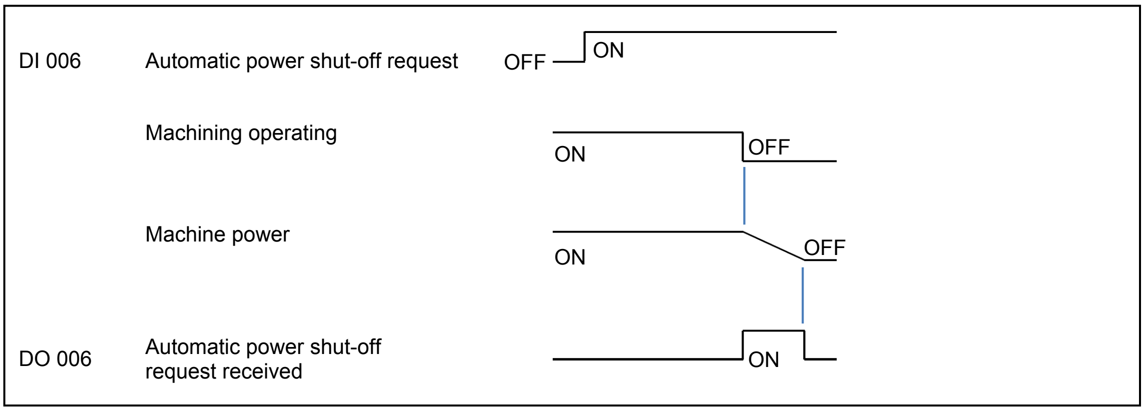


Fig. 6-8 Automatic power shut-off request signal

10. DI 007: (No allocation)

11. DI 008 (DI 010, DI 012, DI 014): Fixture # clamp command

A. Function

Fixtures on the machine can be clamped from the robot side using these signals.

DI 008: Fixture 1

DI 010: Fixture 2

DI 012: Fixture 3

DI 014: Fixture 4

Note 1: Fixtures are optional specifications of the machine.

Note 2: Signals DI 012 and DI 014 can be used for Machine Type-A.

B. Input conditions and operation

Turning any of these signals ON clamps the corresponding fixture.

The signal needs to remain ON until clamping of the fixture has completed.

Whether clamping of the fixture has completed or not can be checked using the corresponding "DO 008/DO 010/DO 012/DO 014: Fixture # clamp complete signal" signal.

Turn the signal OFF after detecting turning ON of the corresponding DO 008/DO 010/DO 012/DO 014 signal.

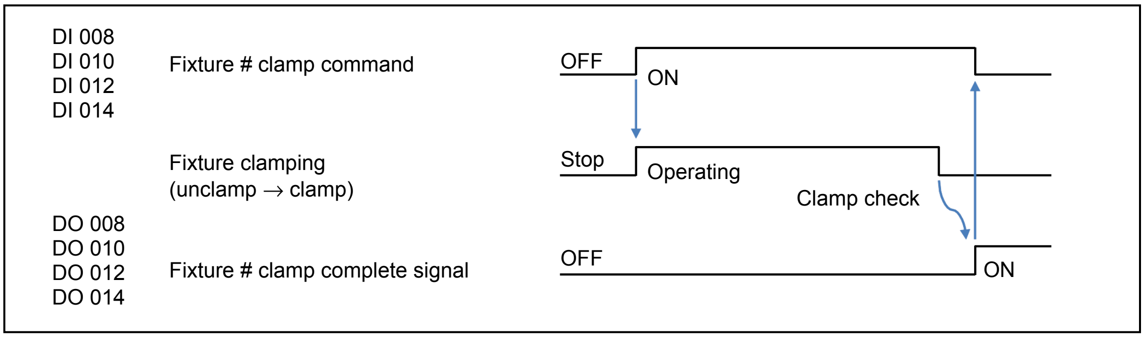


Fig. 6-9 Fixture # clamp signal

12. DI 009 (DI 011, DI 013, DI 015): Fixture # unclamp command

A. Function

Fixtures on the machine can be unclamped from the robot side using these signals.

DI 009: Fixture 1

DI 011: Fixture 2

DI 013: Fixture 3

DI 015: Fixture 4

Note 1: Fixtures are optional specifications of the machine.

Note 2: Signals DI 013 and DI 015 can be used for Machine Type-A.

B. Input conditions and operation

Turning any of these signals ON unclamps the corresponding fixture.

The signal needs to remain ON until unclamping of the fixture has completed.

Whether unclamping of the fixture has completed or not can be checked using the corresponding "DO 009/DO 011/DO 013/DO 015: Fixture # unclamp complete signal" signal.

Turn the signal OFF after detecting turning ON of the corresponding DO 009/DO 011/DO 013/DO 015 signal.

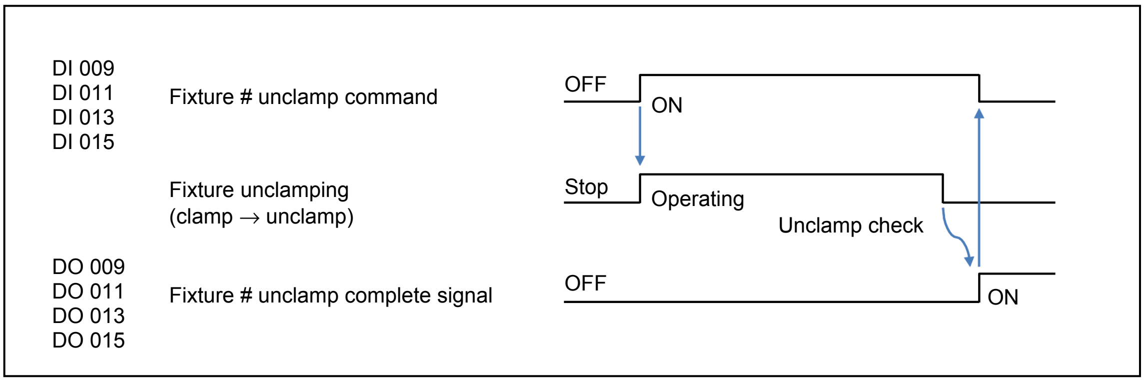


Fig. 6-10 Fixture # unclamp signal

13. DI 100: Target work number data

A. Function

The work number can be specified and set at the machine using this signal.

(Work number: Program number to be executed on a machining center)

B. Input conditions and operation

- Field network communication

  Specify the work number using 32-byte ASCII code data.

  Set the target work number before turning the "DI 101: Work number search start" signal ON.

Example : Work number = ABC123

0	Bit 0 to 7	0 × 33	= 3
1	Bit 8 to F	0 × 32	= 2
2	Bit 0 to 7	0 × 33	= 1
3	Bit 8 to F	0 × 43	= C
4	Bit 0 to 7	0 × 42	= B
5	Bit 8 to F	0 × 41	= A
6	Bit 0 to 7	0 × 00	= null
⋮			
15	Bit 8 to F	0 × 00	= null

16	Bit 0 to 7	0 × 00	= null
17	Bit 8 to F	0 × 00	= null
⋮			= null
			= null
32	Bit 8 to F	0 × 00	= null

14. DI 101: Work number search start

A. Function

Used to start the work number search function using the set work number.

B. Input conditions and operation

Turning this signal ON starts the work number search function at the machine.

It is necessary to set "DI 100: Target work number data" beforehand.

Whether the work number search has completed or not can be checked using the corresponding "DO 101: Work number search complete" signal.

The signal needs to remain ON until the work number search has completed.

Turn the signal OFF after detecting turning ON of the DO 101 signal.

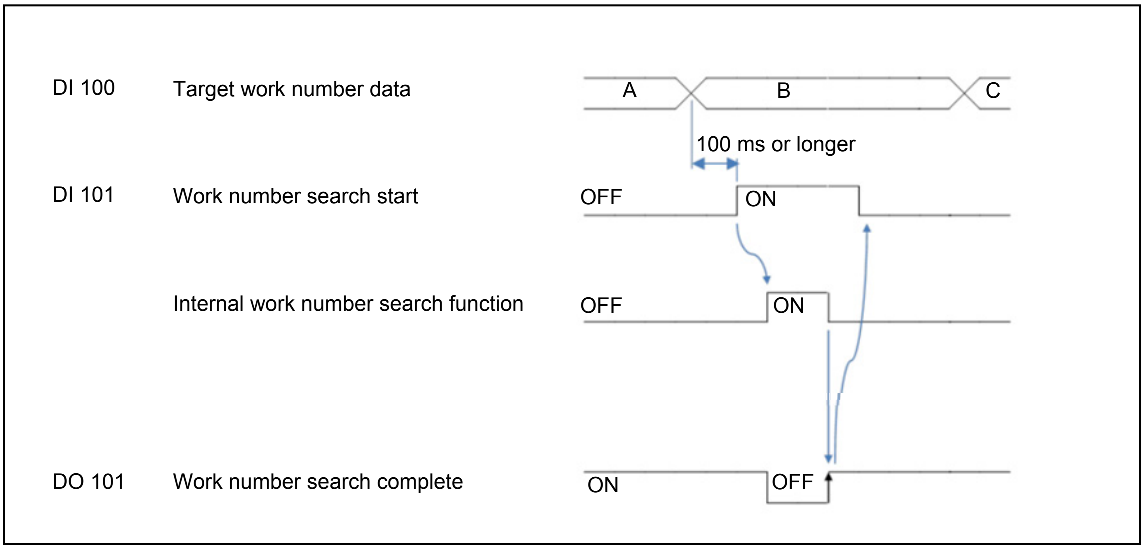


Fig. 6-11 Work number search start signal

15. DI 102: Cycle start command

A. Function

Used to output a cycle start command to the machine from the robot side.

The machine starts a machining cycle according to this signal.

B. Input conditions and operation

This signal is enabled only while the “DO 102: Cycle start permission signal” signal is ON.

It is necessary to check the status of DO 102 before turning this signal ON.

Turning this signal ON outputs a cycle start command to the machine.

The operation status of the machine can be checked using the “DO 103: Machine operating” signal.

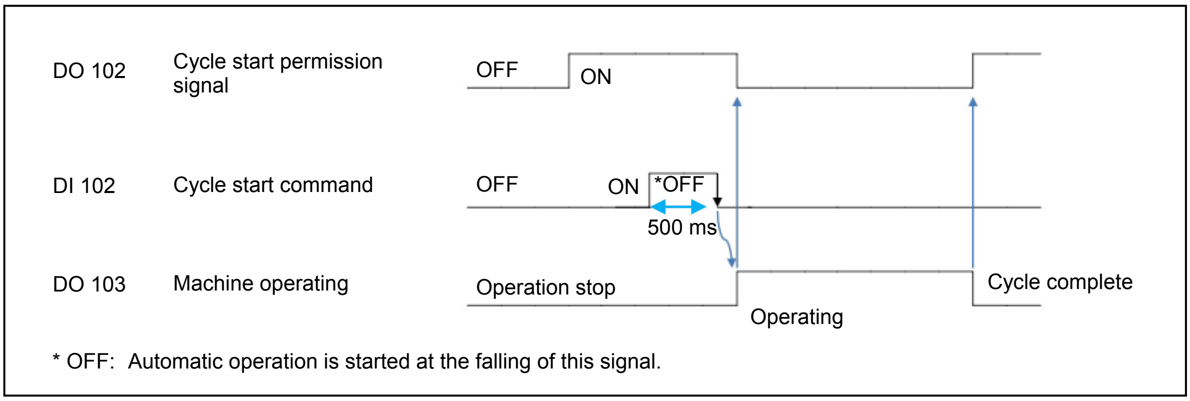


Fig. 6-12 Cycle start command signal

16. DI 103: NC reset

A. Function

The NC unit can be reset and initialized using this signal.

The NC unit of the machine will be reset with the machining cycle immediately stopped.

B. Input conditions and operation

Turning this signal ON resets the NC unit with the machining cycle being executed stopped immediately.

Use pulse signals of around 0.5 s to 1 s for this signal.

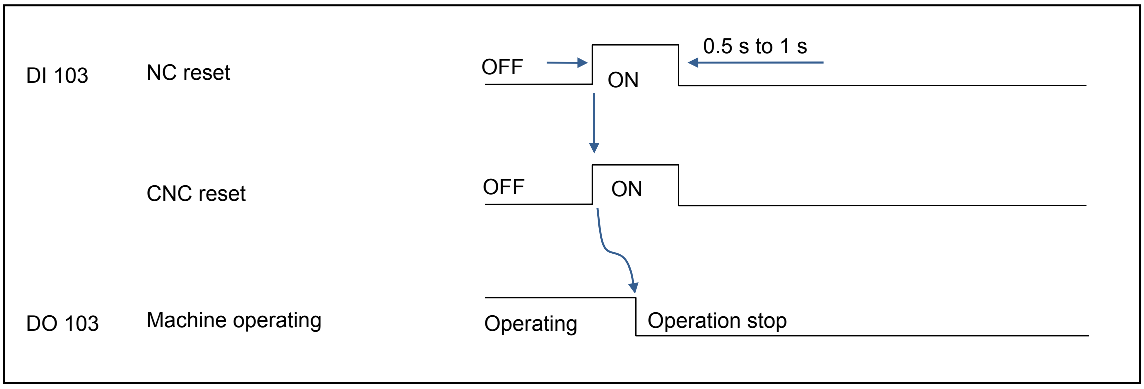


Fig. 6-13 NC reset signal

17. DI 104: All machining complete

A. Function

Used to notify the machine that all the machining has completed.

The machining cycle can be ended from the robot side using this signal.

B. Input conditions and operation

Set this signal at the end of the robot cycle.

The machining cycle at the machine side can be ended from the robot side by turning this signal ON while the machine is waiting for the completion of a robot service M code.

Note : This signal enables ending of a machining cycle where machining processes are looped, by M99 for example.

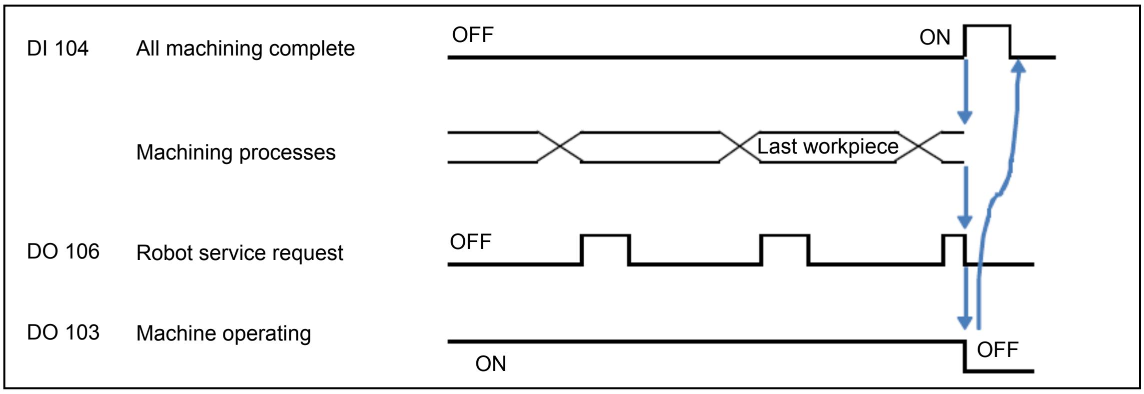


Fig. 6-14 All machining complete signal

18. DI 105: (No allocation)

19. DI 106: Robot Service Finished Signal

A. Function

This complete signal is used in response to the robot service call from the machine.

The signal needs to remain ON until the robot service sequence has completed.

B. Input conditions and operation

Turn the signal ON upon completion of the robot service.

The machining program of the machine is suspended at the robot service M code until this signal is turned ON.

The service call request can be checked using the “DO 106: Robot service request” signal.

The signal needs to remain ON until DO 106 has turned OFF.

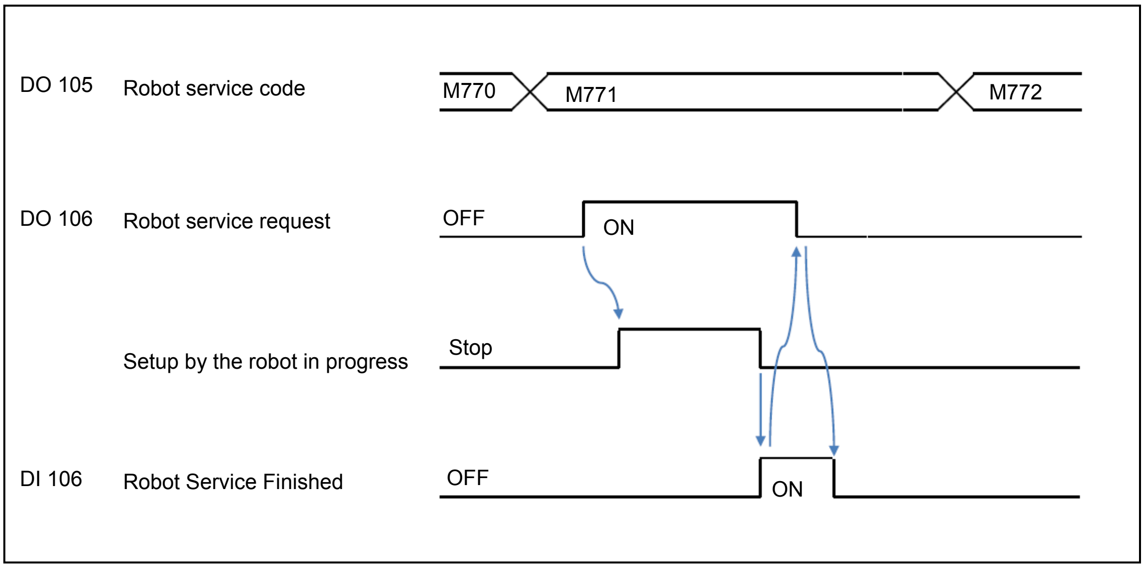


Fig. 6-15 Robot Service Finished Signal

20. DI 107: Machine front door / side door open command

A. Function

The front / side door of the machine can be opened using this signal.

B. Input conditions and operation

Turning this signal ON opens the front / side door of the machine.

The signal needs to remain ON until the front / side door is fully open.

The opening status of the front / side door can be checked using the “DO 107: Front / Side door open complete signal” signal.

Turn the signal OFF after detecting turning ON of the DO 107 signal.

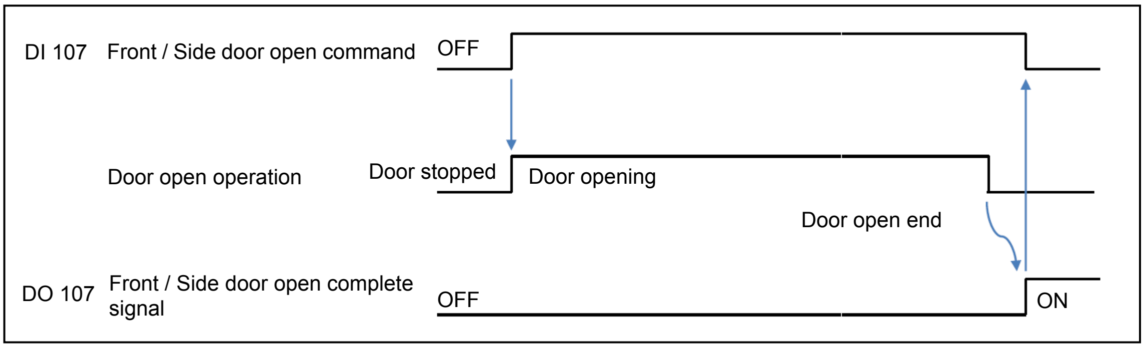


Fig. 6-16 Front / Side door open command signal

21. DI 108: Machine front / side door close command

A. Function

The front / side door of the machine can be closed using this signal.

B. Input conditions and operation

Turning this signal ON closes the front / side door of the machine.

The signal needs to remain ON until the front / side door is fully closed.

The closing status of the front / side door can be checked using the “DO 108: Front / Side door close complete signal” signal.

Turn the signal OFF after detecting turning ON of the DO 108 signal.

Note that the front / side door close operation is disabled when “DI 109: Robot clear” is OFF.

Remark : For details, refer to the next item.

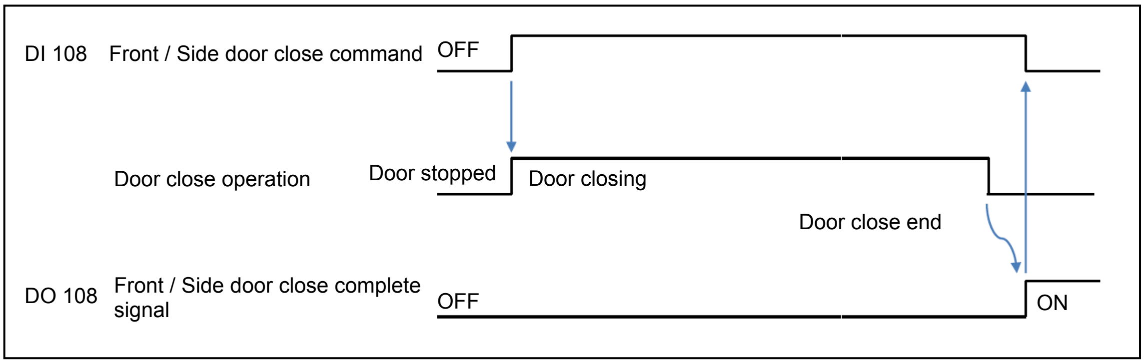


Fig. 6-17 Front / Side door close signal

22. DI 109: Robot clear

A. Function

Turning this signal ON enables closing of the front / side door of the machine.

Turning this signal OFF disables closing of the front / side door of the machine.

B. How to use

This signal is used for the interlocking of the front / side door of the machine.

C. Input conditions and operation

Turn the signal OFF while the robot is inside the machining area of the machine.

Turn the signal ON while the robot is outside the machining area of the machine.

23. DI 110 - DI 129: (No allocation)

6-2 Details on Output Signals (Mazak → Robot)

1. SO 000: Emergency stop signal OUTPUT

A. Function

Used to notify the emergency stop status of the machine to the robot side.

Note : This signal is provided using a double dry contact (no-voltage contact).

- For details, refer to 4-4-1 “Safety signals”.

B. How to use

The robot can monitor the emergency stop status of the machine using this signal.

This signal is used to check for an emergency stop input.

C. Output conditions

This signal turns OFF (open) under the following conditions.

- The emergency stop input is turned OFF.
- The emergency stop button is pressed.
- The control power of the machine is turned off.

2. SO 001: Side door closed

A. Function

Used to notify the status of the side door to the robot side.

Note : This signal is provided using a double dry contact (no-voltage contact).

- For details, refer to 4-4-1 “Safety signals”.

B. How to use

The robot system can check the open/close status of the side door of the machine using this signal.

At the robot system, judge the emergency stop input conditions (SI 000) according to the side door status.

C. Output conditions

This signal turns OFF (open) under the following conditions.

- The side door of the machine is not closed.
- The side door is not locked.
- The control power of the machine is turned off.

3. SO 001: Front door closed

A. Function

Used to notify the status of the front door to the robot side.

Note : This signal is provided using a double dry contact (no-voltage contact).

- For details, refer to 4-4-1 “Safety signals”.

B. How to use

The robot system can check the open/close status of the front door of the machine using this signal.

At the robot system, judge the emergency stop input conditions (SI 000) according to the front door status.

C. Output conditions

This signal turns OFF (open) under the following conditions.

- The front door of the machine is not closed.
- The front door is not locked.
- The control power of the machine is turned off.

4. DO 000: Communication check OUTPUT

A. Function

Normal communication between the machine and robot is checked by switching this signal.

The robot can monitor whether the communication is normal using this signal.

If switching of this signal does not occur for a certain period of time, a communication error between the robot and machine is assumed.

B. How to use

This signal is used to detect communication errors.

C. Output conditions

This signal is turned ON when turning OFF of the “DI 000: Communication check INPUT” signal is detected.

This signal is turned OFF when turning ON of the “DI 000: Communication check INPUT” signal is detected.

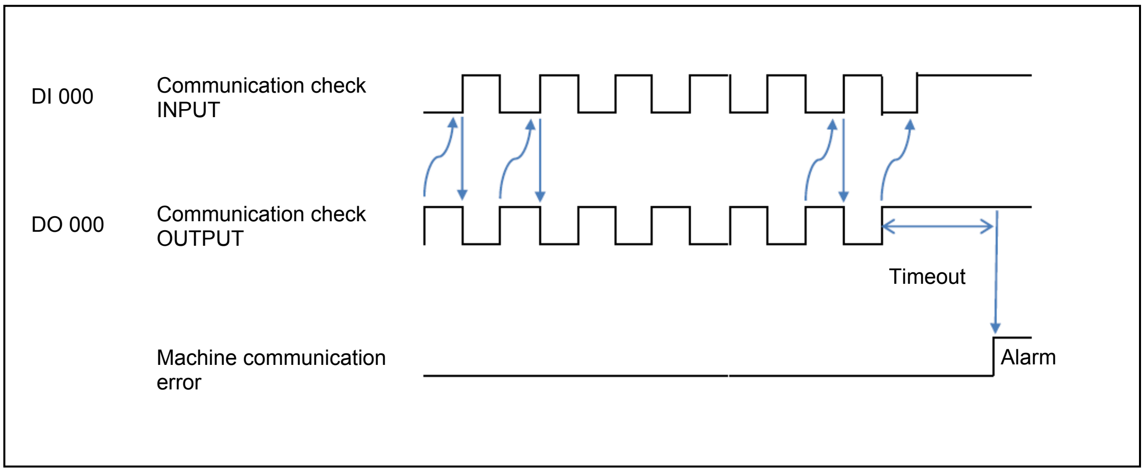


Fig. 6-18 Communication check OUTPUT signal

5. DO 001: Machine ready

A. Function

Used to notify whether the robot interface function is ready for use or not.

B. How to use

This signal is used to check the machine ready status.

C. Output conditions

This signal turns ON when the robot interface function is ready for use.

This signal turns OFF under the following conditions.

- Emergency stop state
- While the robot is offline

6. DO 002: Robot stop request

A. Function

This signal is used to notify the robot operation prohibited status.

B. How to use

This signal needs to be monitored continuously at the robot side.

Stop axis operation of the robot when this signal is turned OFF.

When this signal is turned ON after being turned OFF once, do not recover the axis operation of the robot in response to the rising of the signal.

C. Output conditions

This signal is ON in the normal status.

This signal turns OFF under the following conditions.

- Turning OFF of the safety fence close signal (SI 001) is detected.

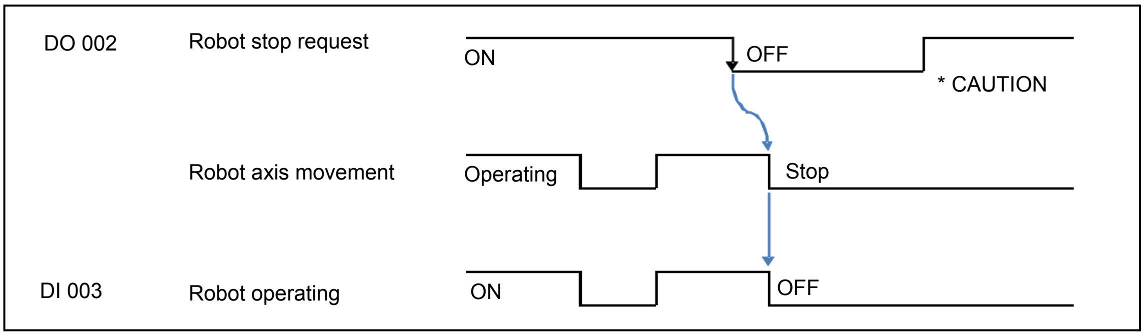


Fig. 6-19 Robot stop request signal

Note : Do not recover robot operation in response to the rising of the signal.

7. DO 003: Machine operating panel at retract position

This signal is normally ON. Check that the robot operation will not cause interference with the machine operating panel before starting robot operation.

8. DO 004: Machine alarm status

A. Function

Used to notify that the machine is in an alarm state.

B. How to use

This signal is used to perform necessary checks on the machine status.

C. Output conditions

This signal turns ON when there is a condition that may hinder robot operation at the machine.

Example :

- Actuator timeout (front / side door)
- General error (machining program error/operation error)

This signal turns OFF when the alarm state is cleared.

9. DO 005: (No allocation)

10. DO 006: Automatic power shut-off request received

A. Function

Used to notify that the automatic power shut-off request from the robot has been accepted.

B. How to use

This signal is used as the confirmation signal in response to the “DI 006: Automatic power shut-off request” signal.

It is also used as the trigger at the robot side for shutting the power off.

C. Output conditions

If the DI 006 signal is turned ON during machine operation, the DO 006 signal turns ON when automatic operation of the machine has stopped.

The signal turns OFF upon turning the power off.

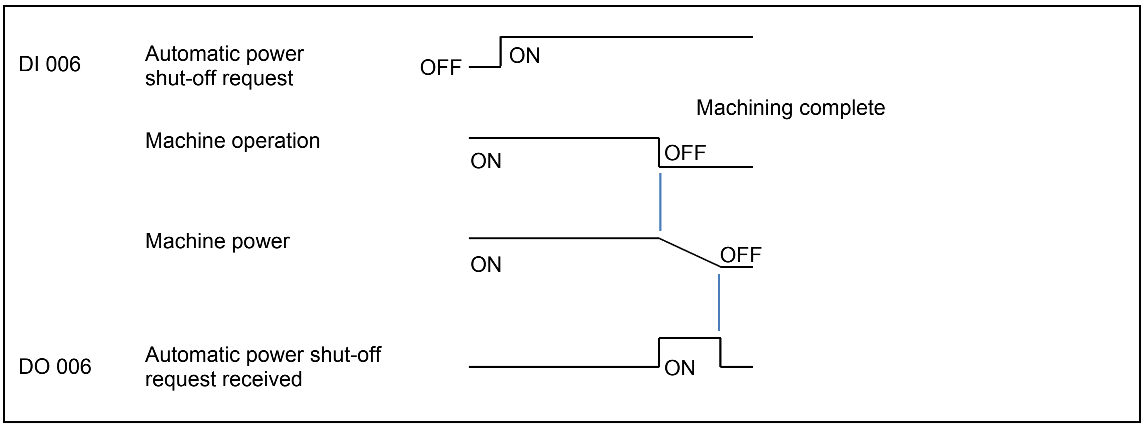


Fig. 6-20 Automatic power shut-off request received signal

11. DO 007: (No allocation)

12. DO 008 (DO010, DO012, DO014): Fixture # clamp complete signal

A. Function

Used to notify completion of clamping operation of the corresponding fixture to the robot side.

DO 008: Fixture 1

DO 010: Fixture 2

DO 012: Fixture 3

DO 014: Fixture 4

Note 1: Fixtures are optional specifications of the machine.

B. How to use

These signals are used as the confirmation signals in response to the corresponding “DI 008/DI 010/DI 012/DI 014: Fixture clamp command” signal.

C. Output conditions

The signal turns ON upon completion of clamping operation of the corresponding fixture.

The clamp state of the fixtures is established by the elapse of the preset time.

The signal turns OFF when the fixture starts unclamping.

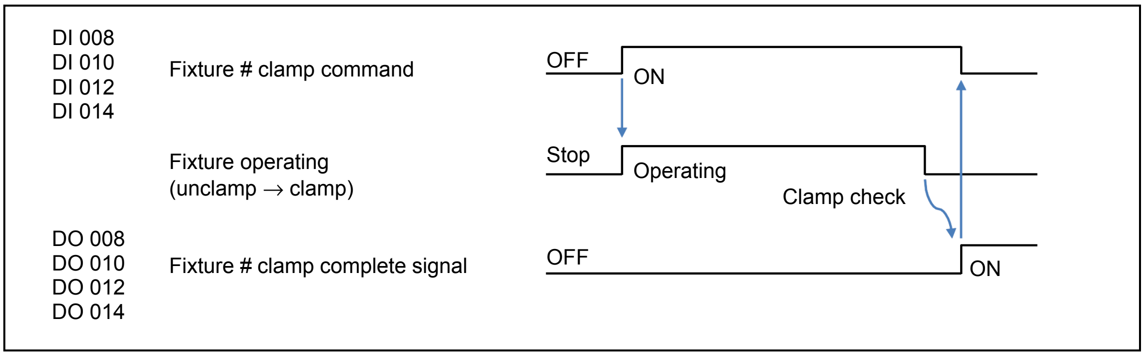


Fig. 6-21 Fixture # clamp signal

13. DO 009 (DO 011, DO013, DO015): Fixture # unclamp complete signal

A. Function

Used to notify completion of unclamping operation of the corresponding fixture to the robot side.

DO 009: Fixture 1

DO 011: Fixture 2

DO 013: Fixture 3

DO 015: Fixture 4

Note 1: Fixtures are optional specifications of the machine.

B. How to use

These signals are used as the confirmation signals in response to the corresponding “DI 009/DI 011/DI 013/DI 015: Fixture # unclamp command” signal.

C. Output conditions

The signal turns ON upon completion of unclamping operation of the corresponding fixture.

The unclamp state of the fixtures is established by the elapse of the preset time.

The signal turns OFF when the fixture starts clamping.

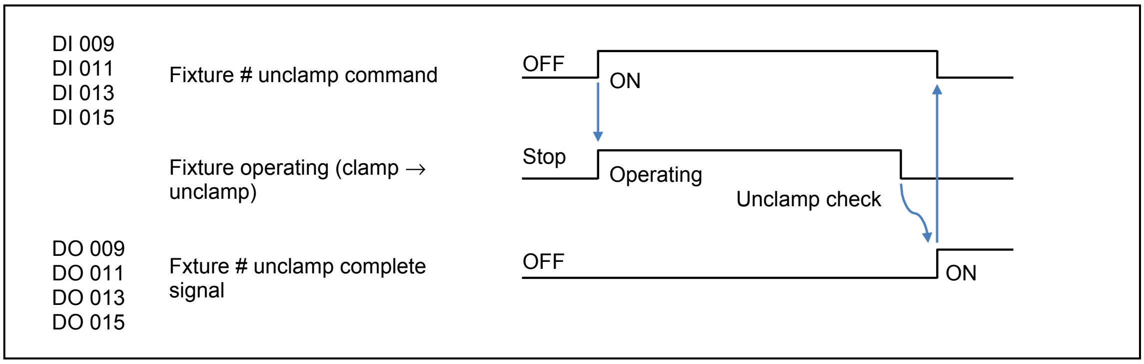


Fig. 6-22 Fixture # unclamp signal

14. DO 016 (DO 017): (No Allocation)

15. DO 100: Current work number

A. Function

This data is used to notify the current work number.

B. How to use

This data is used to check the current work number.

C. Output conditions

The current work number is set by outputting this data.

This data does not change while the DO 101 signal is ON.

Define the work number with a 32-byte ASCII code.

Example : Work number = ABC123

0	Bit 0 to 7	0 × 33	= 3
1	Bit 8 to F	0 × 32	= 2
2	Bit 0 to 7	0 × 33	= 1
3	Bit 8 to F	0 × 43	= C
4	Bit 0 to 7	0 × 42	= B
5	Bit 8 to F	0 × 41	= A
6	Bit 0 to 7	0 × 00	= null
⋮			
15	Bit 8 to F	0 × 00	= null

16	Bit 0 to 7	0 × 00	= null
17	Bit 8 to F	0 × 00	= null
	⋮		= null
			= null
32	Bit 8 to F	0 × 00	= null

16. DO 101: Work number search complete

A. Function

This signal is used to notify the completion work number search function.

B. How to use

The rising of this signal can be used as the confirmation signal in response to the “DI 101: Work number search start” signal.

C. Output conditions

This signal turns ON when the work number search completes.

This signal turns OFF when the work number search function starts.

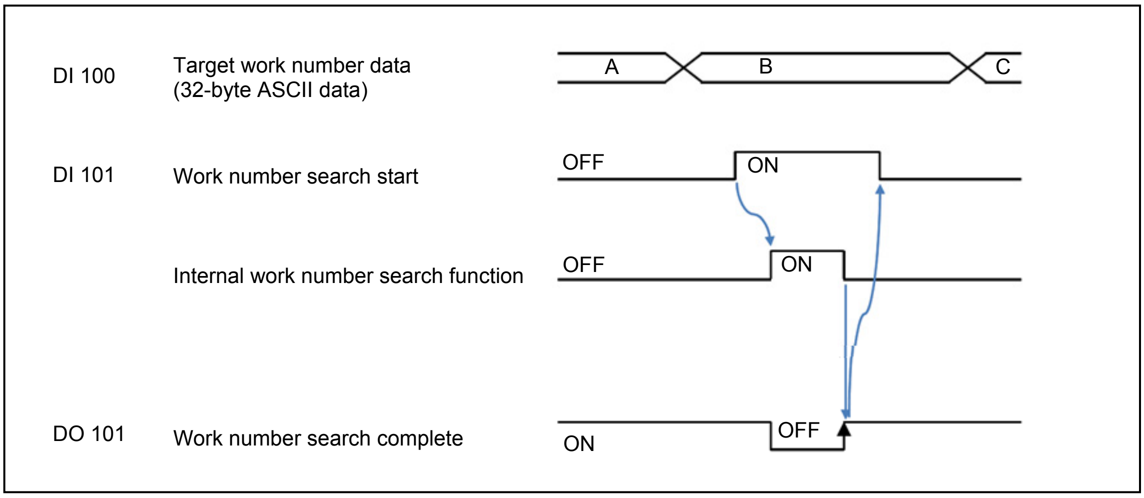


Fig. 6-23 Work number search complete signal

17. DO 102: Cycle start permission signal

A. Function

Used to notify that cycle start can be specified.

B. How to use

The robot side can check that cycle start on the machine is possible using this signal.

C. Output conditions

This signal turns ON when cycle start is possible on the machine.

This signal turns OFF under the following conditions.

- Automatic operation in progress on the machine
- Cycle start not possible on the machine

Example :

- An inappropriate state (manual mode/MDI mode/setup mode) is selected.
- The front / side door of the machine is not closed.

18. DO 103: Machine operating

A. Function

This signal is used to notify that the cycle start is in progress on the machine.

B. How to use

This signal is used as the confirmation signal in response to the “DI 102: Cycle start command” signal.

C. Output conditions

This signal turns ON during automatic operation.

This signal turns OFF when automatic operation has completed.

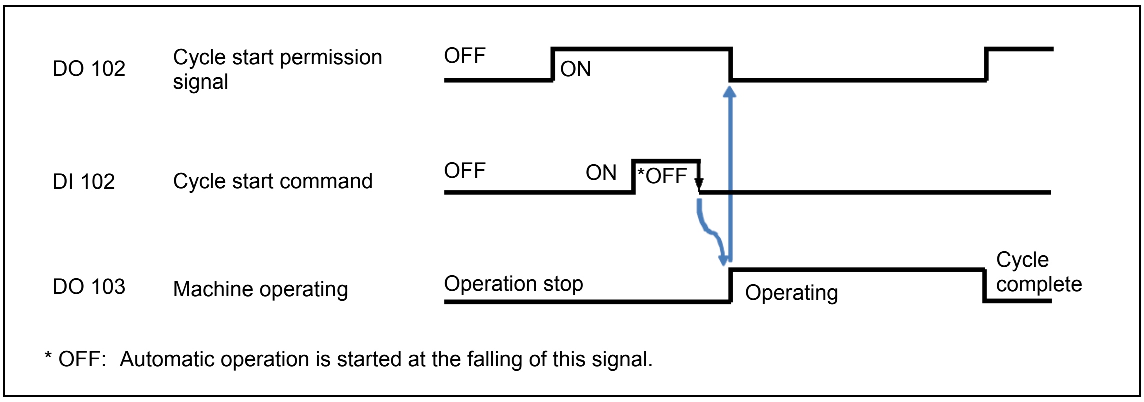


Fig. 6-24 Machine operating signal

19. DO 104: Machining complete

A. Function

This signal is used to notify the completion of machining cycle.

B. How to use

When robot service M codes are not used, this signal is used as the trigger for starting robot service.

C. Output conditions

This signal turns ON when the machining cycle called by M30/M02/M999 has completed.

This signal turns OFF when a machining cycle starts.

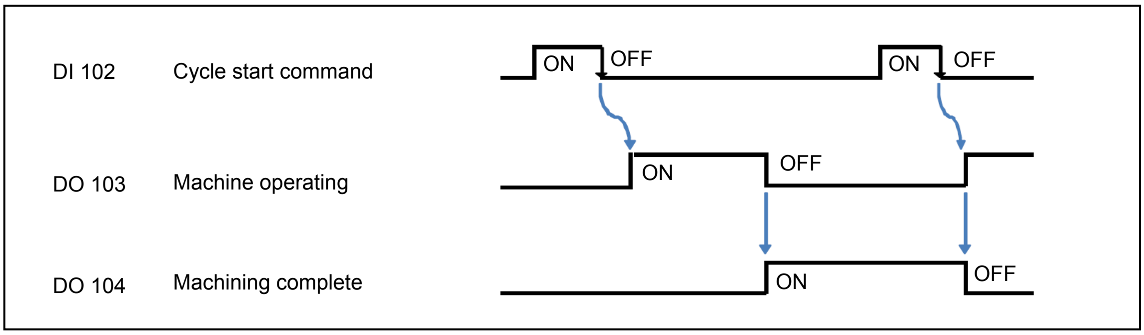


Fig. 6-25 Machining complete signal

20. DO 105: Robot service code

A. Function

This data is used to notify the command to be executed by a robot service request.

B. How to use

Read this data when the “DO 106: Robot service request” signal is output.

C. Output conditions

This data does not change while the “DO 106: Robot service request” signal is ON.

The machine outputs commands corresponding to M770 to M779 (a total of 10 M codes).

The following shows the data format. (Field network: 1 byte)

M Code	Field Network
M770	0 × 00
M771	0 × 01
M772	0 × 02
⋮	⋮
M778	0 × 08
M779	0 × 09

21. DO 106: Robot service request

A. Function

This signal is used to notify a workpiece processing request from the machine side.

B. How to use

This signal needs to be monitored continuously at the robot side.

Start workpiece processing at the robot side when this signal is ON.

C. Output conditions

This signal turns ON when executing M770 to M779 in the machining program.

The corresponding command is set for the “DO 105: Robot service code” signal before turning this signal ON.

This signal turns OFF when the “DI 106: Service call complete” signal is input.

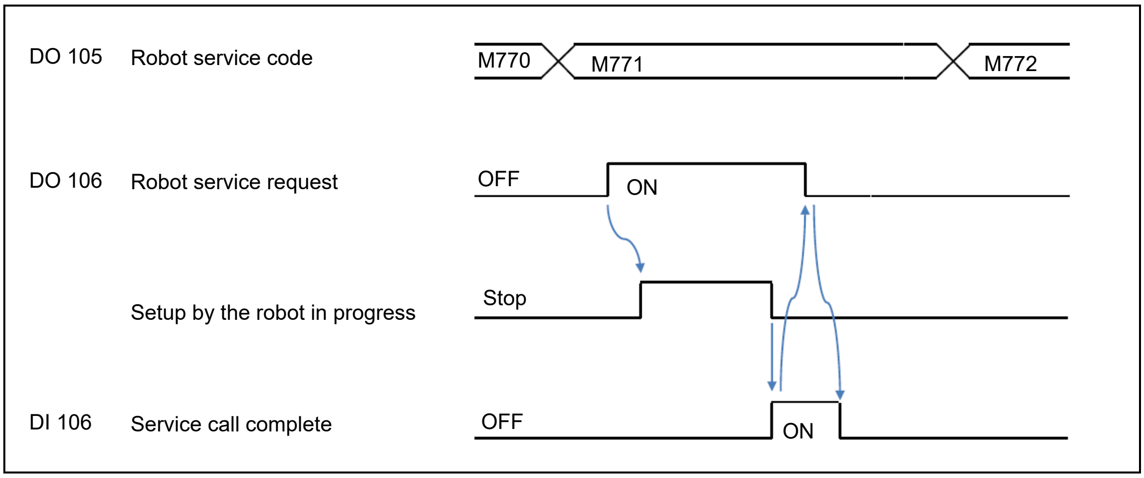


Fig. 6-26 Robot service request signal

22. DO 107: Machine front / side door open complete signal

A. Function

Used to notify that the front / side door of the machine has opened.

B. How to use

This signal is used as the confirmation signal in response to the “DI 107: Front / Side door open command” signal.

C. Output conditions

This signal turns ON when the front / side door has opened.

This signal turns OFF when the front / side door is not open.

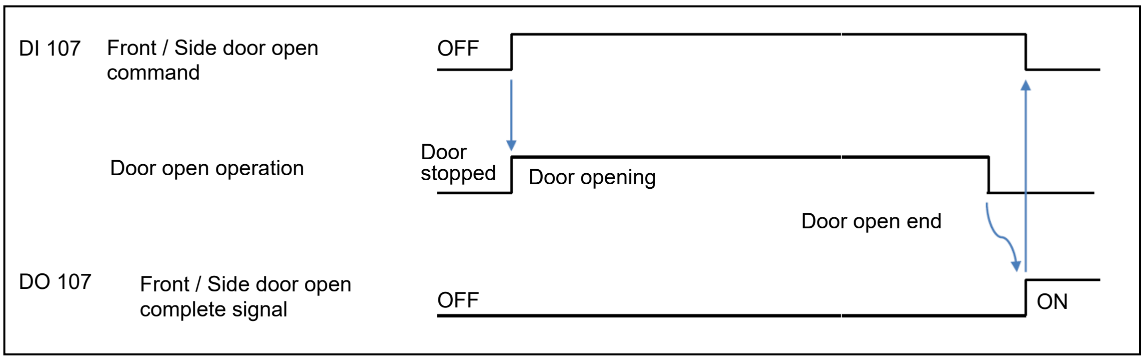


Fig. 6-27 Front / Side door open complete signal

23. DO 108: Machine front / side door close complete signal

A. Function

Used to notify that the front / side door of the machine is closed.

B. How to use

This signal is used as the confirmation signal in response to the “DI 108: Front / Side door close command” signal.

C. Output conditions

This signal turns ON when the front / side door is closed.

This signal turns OFF when the front / side door is not closed.

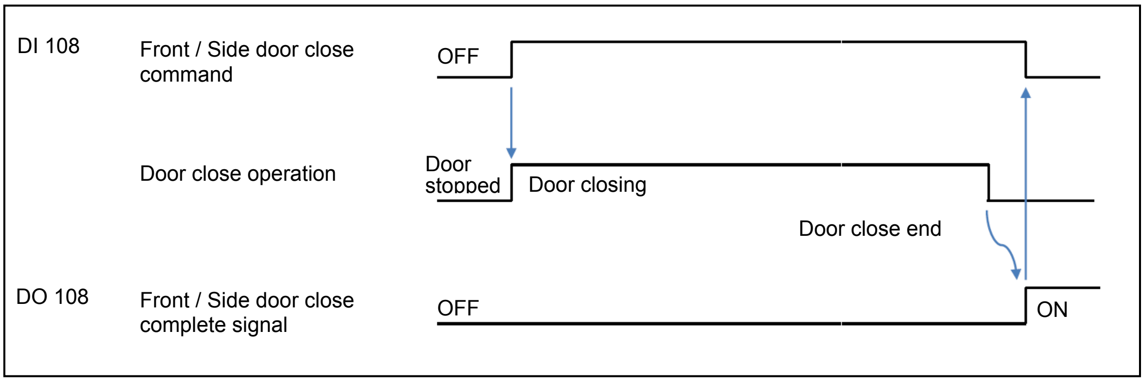


Fig. 6-28 Front / Side door close complete signal

24. DO 109: Robot access to machine permission

A. Function

Used to notify that the robot can enter the machining area of the machine.

B. How to use

The robot side needs to check that this signal is ON before entering into the machining area of the machine.

C. Output conditions

This signal turns ON when the machine is ready for entry of the robot.

This occurs when all of the following conditions are satisfied.

a. The front / side door is open.

b. All the axes are at a stop.

c. The spindle is at a stop.

d. The Z-axis is at the zero point and the X- and Y-axis are at their setup positions.

e. The 4th- and 5th-axis are at an access permitted position. (Only when equipped with the corresponding additional axis options)

This signal turns OFF if any of the conditions above is not satisfied.

25. DO 110 - DO 129: (No allocation)

7 REGISTER ADDRESS MAP FOR FIELD NETWORK COMMUNICATION

7-1 Ethernet IP, PROFIBUS-DP, CC-Link

7-1-1 Input data table

The following shows the address map for input bit registers.

Areas of 44 bytes need to be secured for the input bit register addresses.

Note that the address map given here shows the addresses used with the following RS parameter setting.

- RS224 CH1: 16386 (CH2: 25086)

Address		Data Size	Signal No.	Signal Name	Remarks
R16384 (CH1) R25084 (CH2) to R16385 (CH1) R25085 (CH2)		4 bytes	–	Ethernet/IP occupied area	CIP counter data
R16386 (CH1) R25086 (CH2)	0	2 Bit	DI 000	Communication check input	
	1		DI 001	Robot ready	
	2		DI 002	Machine stop request	
	3		DI 003	Robot operating	
	4		DI 004	Robot alarm status	
	5		DI 005	Operator interruption request	
	6		DI 006	Automatic power shut-off request	
	7		–	(No allocation)	
	8		DI 008	Fixture 1 clamp command	Option
	9		DI 009	Fixture 1 unclamp command	
	A		DI 010	Fixture 2 clamp command	
	B		DI 011	Fixture 2 unclamp command	
	C		DI 012	Fixture 3 clamp command	
	D		DI 013	Fixture 3 unclamp command	
	E		DI 014	Fixture 4 clamp command	
	F		DI 015	Fixture 4 unclamp command	
R16387	0	Bit	DI 016	Workpiece seating detection request 1	
	1		DI 017	Workpiece seating detection request 2	
	2		DI 018	Ignore workpiece seating detection alarm 1	
	3		DI 019	Ignore workpiece seating detection alarm 2	
	4		–	(No allocation)	
	5				
	6				
	7				
R16388 to R16390		6 bytes	–	(No allocation)	
R16391 to R16406		32 bytes	DI 100	Target work number data	

Address		Data Size	Signal No.	Signal Name	Remarks
R16392 (CH1) R25092 (CH2)	0	Bit	DI 101	Work number search start	
	1		DI 102	Cycle start command	
	2		DI 103	NC reset	
	3		DI 104	All machining complete	
	4		–	(No allocation)	
	5		DI 106	Robot Service Finished Signal	
	6		DI 107	Machine Front / Side door open command	
	7		DI 108	Machine Front / Side door close command	
	8		DI 109	Robot clear	
R16393 to R16407		30 bytes	–	(No allocation)	
Total		74 bytes			

The following shows the address map for input word registers.

Areas of 32 bytes need to be secured for the input word register addresses.

Note that the address map given here shows the addresses used with the following RS parameter setting.

- RS226 CH1: 16408 (CH2: 25108)

Address	Data Size	Signal No.	Signal Name	Remarks
R16408 (CH1) R25108 (CH2) to R16423 (CH1) R25123 (CH2)	32 bytes	DI 100	Target work number data	
Total	32 bytes			

7-1-2 Output data table

The following shows the address map for output bit registers.

Areas of 44 bytes need to be secured for the output bit register addresses.

Note that the address map given here shows the addresses used with the following RS parameter setting.

- RS225 CH1: 18178 (CH2: 26878)

<table>
<tr><th colspan="2">Address</th><th>Data Size</th><th>Signal No.</th><th>Signal Name</th><th>Remarks</th></tr>
<tr><td colspan="2">R18176 (CH1)
R26876 (CH2)
to
R18177 (CH1)
R26877 (CH2)</td><td>4 bytes</td><td>–</td><td>Ethernet/IP occupied area</td><td>CIP counter data</td></tr>
<tr><td rowspan="16">R18178 (CH1)
R26878 (CH2)</td><td>0</td><td rowspan="16">2 Bit</td><td>DO 000</td><td>Communication check output</td><td></td></tr>
<tr><td>1</td><td>DO 001</td><td>Machine ready</td><td></td></tr>
<tr><td>2</td><td>DO 002</td><td>Robot stop request</td><td></td></tr>
<tr><td>3</td><td>DO 003</td><td>Machine operating panel at retract position</td><td></td></tr>
<tr><td>4</td><td>DO 004</td><td>Machine alarm status</td><td></td></tr>
<tr><td>5</td><td>–</td><td>(No allocation)</td><td></td></tr>
<tr><td>6</td><td>DO 006</td><td>Automatic power shut-off request received</td><td></td></tr>
<tr><td>7</td><td>–</td><td>(No allocation)</td><td></td></tr>
<tr><td>8</td><td>DO 008</td><td>Fixture 1 clamp complete signal</td><td rowspan="4">Option</td></tr>
<tr><td>9</td><td>DO 009</td><td>Fixture 1 unclamp complete signal</td></tr>
<tr><td>A</td><td>DO 010</td><td>Fixture 2 clamp complete signal</td></tr>
<tr><td>B</td><td>DO 011</td><td>Fixture 2 unclamp complete signal</td></tr>
<tr><td>C</td><td>DO 012</td><td>Fixture 3 clamp complete signal</td><td rowspan="4"></td></tr>
<tr><td>D</td><td>DO 013</td><td>Fixture 3 unclamp complete signal</td></tr>
<tr><td>E</td><td>DO 014</td><td>Fixture 4 clamp complete signal</td></tr>
<tr><td>F</td><td>DO 015</td><td>Fixture 4 unclamp complete signal</td></tr>
<tr><td rowspan="8">R18179</td><td>0</td><td rowspan="8">Bit</td><td>DO 016</td><td>Seating detection complete signal 1</td><td rowspan="8"></td></tr>
<tr><td>1</td><td>DO 017</td><td>Seating detection complete signal 2</td></tr>
<tr><td>2</td><td>DO 018</td><td>Workpiece seating detection alarm 1</td></tr>
<tr><td>3</td><td>DO 019</td><td>Workpiece seating detection alarm 2</td></tr>
<tr><td>4</td><td rowspan="4">–</td><td rowspan="4">(No allocation)</td></tr>
<tr><td>5</td></tr>
<tr><td>6</td></tr>
<tr><td>7</td></tr>
<tr><td colspan="2">R18180
to
R18182</td><td>6 bytes</td><td>–</td><td>(No allocation)</td><td></td></tr>
<tr><td colspan="2">R18183
to
R18198</td><td>32 bytes</td><td>DO 100</td><td>Current work number</td><td></td></tr>
</table>

Address		Data Size	Signal No.	Signal Name	Remarks
R18184 (CH1) R26884 (CH2)	0	Bit	DO 101	Work number search complete	
	1		DO 102	Cycle start permission signal	
	2		DO 103	Machine operating	
	3		DO 104	Machining complete	
	4		DO 105	Robot service code 0	
	5			Robot service code 1	
	6			Robot service code 2	
	7			Robot service code 3	
	8		DO 106	Robot service request	
	9		DO 107	Machine Front / Side door open finished signal	
	A		DO 108	Machine Front / Side door close finished signal	
	B		DO 109	Robot access to machine permitted	
	C		DO 110	Robot Service Finished Confirmation	
R18185 to R18199		30 bytes	–	(No allocation)	
Total		74 bytes			

The following shows the address map for output word registers.

Areas of 32 bytes need to be secured for the output word register addresses.

Note that the address map given here shows the addresses used with the following RS parameter setting.

- RS227 CH1: 18200 (CH2: 26900)

Address	Data Size	Signal No.	Signal Name	Remarks
R18200 (CH1) R26900 (CH2) to R18215 (CH1) R26915 (CH2)	32 bytes	DO 100	Current work number	
Total	32 bytes			

8 APPENDIX

8-1 Connections

8-1-1 How to connect cables between the robot and machine

The machine and robot are connected using cables.

There are two cables, one for a field network and the other for safety-related communication.

The following describes how to connect each cable.

The power supply for the robot needs to be prepared by the customer.

8-1-2 Field network cable

A connector port (X****) for a field network is provided on the NC unit inside the electrical control cabinet of the machine.

Route the cable to the connector port on the NC unit.

Mazak takes responsibility for connection to the connector port on the NC unit, and the external connection from the connector is the customer's responsibility.

Fit Mitsubishi Profibus DP fieldbus unit, FCU8-EX563 to the NC unit A0 in extension slot 3, as shown by the schemes below:

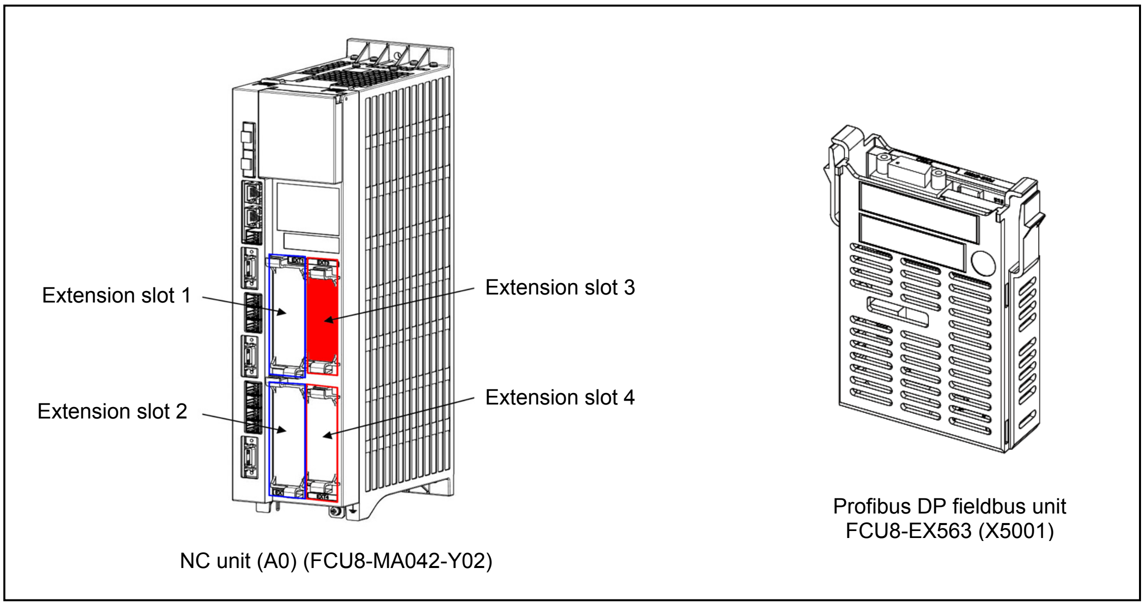


Fig. 8-1 Field network connector port

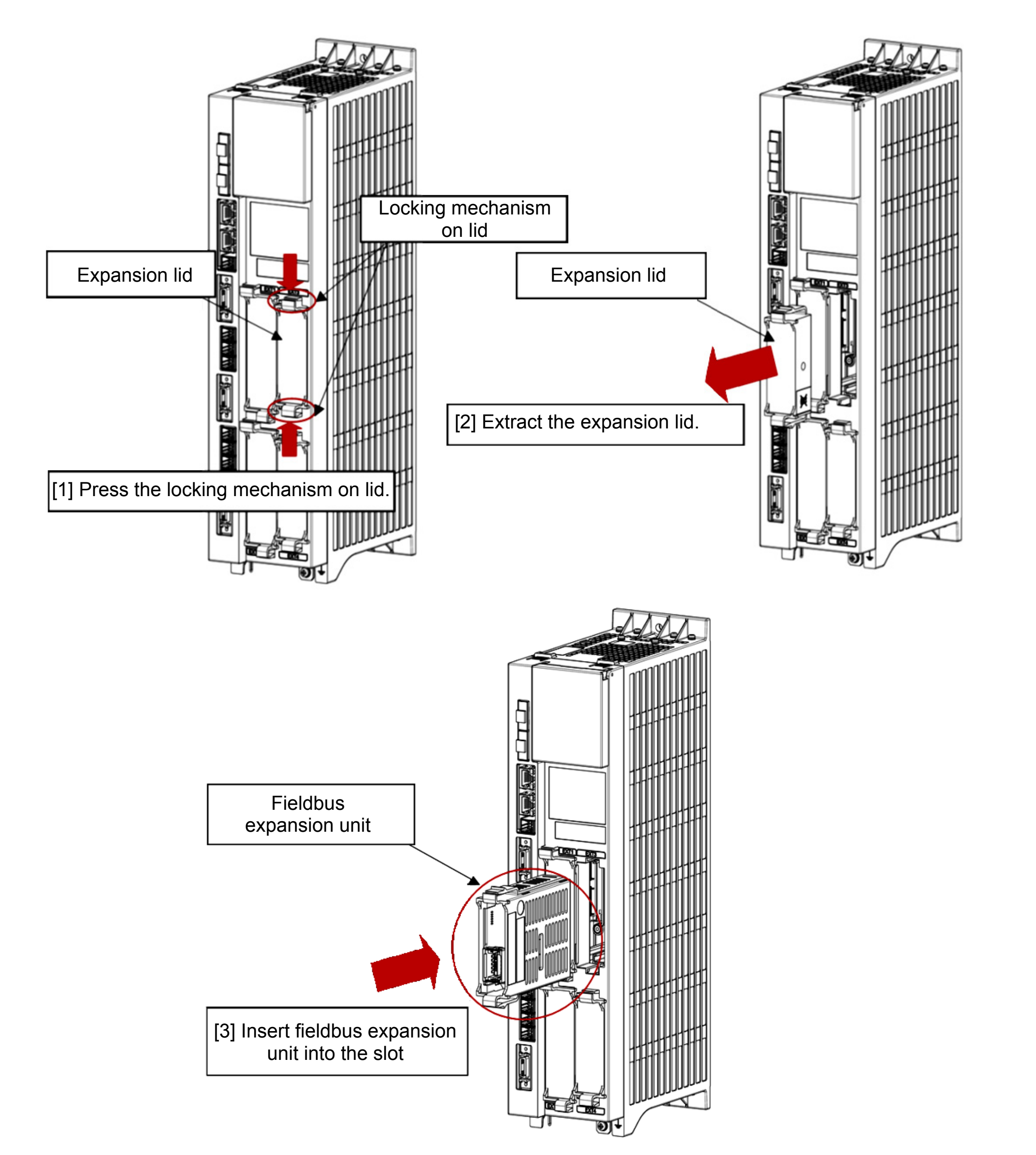


Fig. 8-2 Connecting the fieldbus expansion unit

< Procedure >

(1) Press both ends of the locking mechanism on the expansion lid on the control unit expansion slot 3.

(2) Extract the expansion lid from the control unit.

(3) Insert the expansion unit into slot 3 on the control unit until it clicks into position.

	• The code indicated at "****" differs depending on the machine model. For the device number, refer to the electrical wiring diagrams of the relevant machine.

The cable used for this connection needs to be prepared by the customer.

Use a cable and connectors appropriate for the type of field network.

Connect wires according to the electrical specifications of the signals described in this manual.

8-1-3 Cable for safety-related communication

Connectors for safety signals are provided inside the electrical control cabinet of the machine.

Mazak takes responsibility for connection to these connectors, while the external connection from the connectors is the customer’s responsibility.

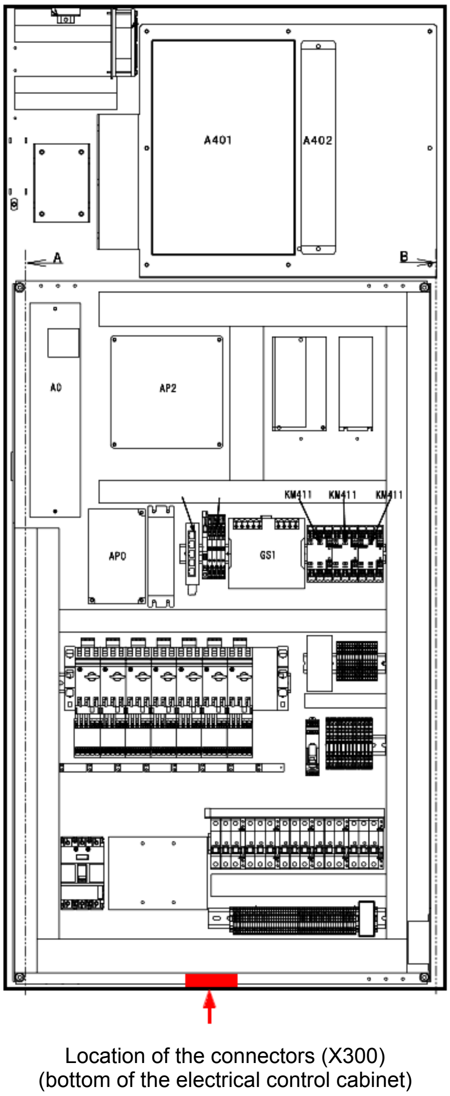


Fig. 8-3 Location of the connectors for safety signals

The cable used for this connection needs to be prepared by the customer.

Connect wires according to the electrical specifications of the signals described in this manual.

8-2 Automatic Operation

8-2-1 Related M codes

Robot services, such as loading of a piece of material and unloading of the completed part, are executed using the following M codes.

The machine requests relevant operations at the robot side using these M codes:

Robot Service Code

M770: Service call 1	M775: Service call 6
M771: Service call 2	M776: Service call 7
M772: Service call 3	M777: Service call 8
M773: Service call 4	M778: Service call 9
M774: Service call 5	M779: Service call 10

When any of the above M codes is specified, a robot service request (DO 106) is output together with the corresponding service code (DO 105) through the robot interface.

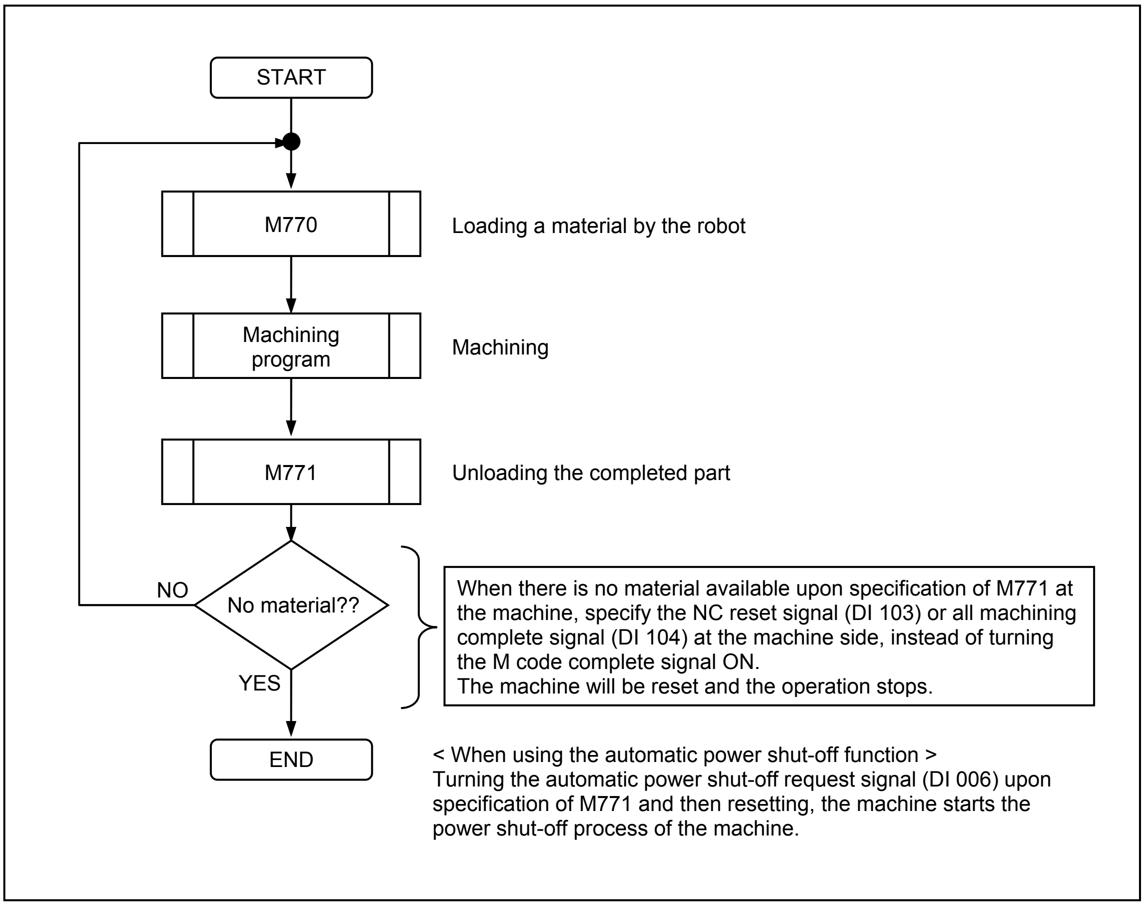
8-2-2 Operation of machining programs

An example operation under the following conditions is described here.

- Workpiece with single process (carrying out machining on the spindle)
- Loading a piece of material onto the spindle and removing the product from the spindle after machining
- M code: M770 Loading of a piece of material by the robot
  M code: M771 Unloading of the machined part by the robot

  (The customer shall decide the details of operations for M codes M770 to M779 by consulting the robot manufacturer.)
- Open and close the front / side door of the machine in accordance with the commands from the robot side during execution of the service calls using M codes (M770/M771).
- To specify operations at the machine side, such as fixture clamping/unclamping, during automatic operation from the robot side, specify the relevant commands after the specification of a service call from the machine.

8-2-3 Flowchart



8-3 Alarm

8-3-1 Structure of the alarm list

This alarm list is written in the following format.

No.	Message	Type of error	Stopped status	Clearing procedure	Display color
[1]	[2] (, ,)	[3]	[4]	[5]	
Cause	[6]				
Action	[7]				

[1] : Alarm number

[2] : Alarm message

[3] : Type of error

Code	Name	Description
A	Operation	A wrong key has been pressed. Or the machine has been operated incorrectly.
B	Registered data	The program or tool data includes an error(s).
C	Servo	Malfunctioning of the servo control mechanism
D	Spindle	Malfunctioning of the spindle control mechanism
E	NC unit	System (hardware/software) error
F	Machine (PLC)	Machine failure
G	External I/O unit	Malfunctioning of external I/O unit

[4] : Stopped status

Code	Status
H	Emergency stop
I	Reset stop
J	Stop in the single block operation mode
K	Feed stop (hold)
L	Operation continued

[5] : Clearing procedure

Code	Procedure
	MAZATROL SmoothX
M	Power off → Eliminate the cause → Power back on
N	Eliminate the cause → Power off → Power back on
O	Eliminate the cause → Press the RESET key on the machine operating panel.
P	Press the RESET key on the machine operating panel.
Q	Eliminate the cause → Tap the button on the NC screen.
S	Tap the button on the NC screen.

[6] : Cause of alarm

[7] : Action to be taken to eliminate the cause

8-3-2 Alarm list

No.	Message	Type of error	Stopped status	Clearing procedure	Display color
1360	ROBOT CONTROLLER MALFUNCTION (, ,)	G	K	O/Q	Red
Cause	There was an abnormality at the robot side.				
Action	There may be an alarm generated at the robot side. Check the robot and eliminate the cause of the alarm. After eliminating the cause, press the RESET key on the machine operating panel or tap the button on the NC screen to clear the alarm.				
1361	COND.MISSING(ROBOT) (, ,)	A/B	K	P/S	Yellow
Cause	Any of the following commands was specified while the robot was inside the machine. • Automatic door close, tool length measuring unit extend, TOOL EYE extend, cycle start				
Action	• Check that the robot online switch is lit. • Check that the memory mode or tape mode is selected. • Check that the front door or safety fence is closed. • Check that the machine setup switch is set to OFF. • Check that R16384 bit 2 (machine stop request) is set to “1”. • Check that R16405 bit 8 [Ethernet IP, PROFIBUS-DP] and R16390 bit 8 [CC-LINK] (robot clear) are set to “1”. After eliminating the cause, press the RESET key on the machine operating panel or tap the button on the NC screen to clear the alarm.				
1362	ROBOT COMMUNICATION ERROR (, ,)	G	K	O/Q	Red
Cause	A communication error between the robot and machine has been detected.				
Action	• Check for loose connectors and cable disconnection. • Check that the robot is communicating. After eliminating the cause, press the RESET key on the machine operating panel or tap the button on the NC screen to clear the alarm.				

8-4 Parameters

<table>
<tr><th>Address</th><th colspan="2">Meaning</th><th>Description</th></tr>
<tr><td rowspan="4">RB125
(bit 1)</td><td colspan="2">Robot interface (advanced)
Communication R register variable specification</td><td rowspan="4">Enables/disables the communication R register variable specification of the robot interface (advanced).
0: Disabled
The communication R register variable specification of the robot interface (advanced) cannot be used.
1: Enabled
The communication R register variable specification of the robot interface (advanced) can be used.

Default setting: 1</td></tr>
<tr><td>Conditions</td><td>Restarting of the machine</td></tr>
<tr><td>Unit</td><td>Bit</td></tr>
<tr><td>Setting range</td><td>0, 1</td></tr>
<tr><td rowspan="4">RB125
(bit 2)</td><td colspan="2">Robot interface (advanced)
Work number I/O method</td><td rowspan="4">Sets the work number I/O method for the robot interface (advanced).
0: Notify text to the NC, starting from its beginning (left end).
[Compatible with conventional method]
1: Notify text to the NC, starting from its end (right end).

Default setting: 1</td></tr>
<tr><td>Conditions</td><td>Restarting of the machine</td></tr>
<tr><td>Unit</td><td>Bit</td></tr>
<tr><td>Setting range</td><td>0, 1</td></tr>
<tr><td rowspan="4">RB163
(bit 0)</td><td colspan="2">Robot interface (advanced)</td><td rowspan="4">Enables/disables the external robot.
0: Disabled
The external robot cannot be used.
1: Enabled
The external robot can be used.

Default setting: 1</td></tr>
<tr><td>Conditions</td><td>Restarting of the machine</td></tr>
<tr><td>Unit</td><td>Bit</td></tr>
<tr><td>Setting range</td><td>0, 1</td></tr>
</table>

<table>
<tr><th>Address</th><th colspan="2">Meaning</th><th>Description</th></tr>
<tr><td rowspan="4">RS224</td><td colspan="2">Initial address of bit data input for the robot interface (advanced)</td><td rowspan="4">Sets the initial R register for the input signal used in field network communications.

Default setting:
• 16384 (CC-Link)
• 16386 (EtherNet/IP, PROFIBUS-DP)
Note :
When 16384 or smaller is set, 16384 is assumed.</td></tr>
<tr><td>Conditions</td><td>Restarting of the machine</td></tr>
<tr><td>Unit</td><td>—</td></tr>
<tr><td>Setting range</td><td>—</td></tr>
<tr><td rowspan="4">RS225</td><td colspan="2">Initial address of bit data output for the robot interface (advanced)</td><td rowspan="4">Sets the initial R register for the output signal used in field network communications.

Default setting:
• 18176 (CC-Link)
• 18178 (EtherNet/IP, PROFIBUS-DP)
Note :
When 18176 or smaller is set, 18176 is assumed.</td></tr>
<tr><td>Conditions</td><td>Restarting of the machine</td></tr>
<tr><td>Unit</td><td>—</td></tr>
<tr><td>Setting range</td><td>—</td></tr>
<tr><td rowspan="4">RS226</td><td colspan="2">Initial address of word data (16-bit) input for the robot interface (advanced)</td><td rowspan="4">Sets the initial R register for the input word data used in field network communications.

Default setting:
16408 (Common to EtherNet/IP, PROFIBUS-DP, CC-Link)
Note :
When 16408 or smaller is set, 16408 is assumed.</td></tr>
<tr><td>Conditions</td><td>Restarting of the machine</td></tr>
<tr><td>Unit</td><td>—</td></tr>
<tr><td>Setting range</td><td>—</td></tr>
<tr><td rowspan="4">RS227</td><td colspan="2">Initial address of word data (16-bit) output for the robot interface (advanced)</td><td rowspan="4">Sets the initial R register for the output word data used in field network communications.

Default setting:
18200 (Common to EtherNet/IP, PROFIBUS-DP, CC-Link)
Note :
When 18200 or smaller is set, 18200 is assumed.</td></tr>
<tr><td>Conditions</td><td>Restarting of the machine</td></tr>
<tr><td>Unit</td><td>—</td></tr>
<tr><td>Setting range</td><td>—</td></tr>
</table>

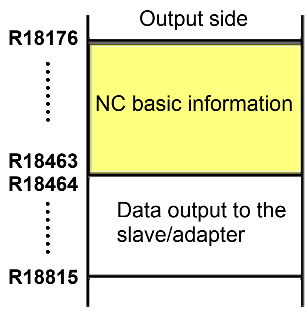
<table>
<tr><th>Address</th><th colspan="2">Meaning</th><th>Description</th></tr>
<tr><td rowspan="4">RL74</td><td colspan="2">4th-axis position to permit robot access to the machine</td><td rowspan="4">Sets the 4th-axis position where the robot can enter inside the machine.
This position is checked as one of the conditions to turn the “DO 109: Robot access to machine permission” signal ON.

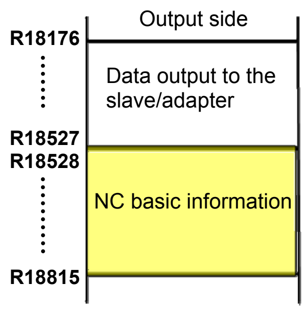
Default setting: 0</td></tr>
<tr><td>Conditions</td><td>Restarting of the machine</td></tr>
<tr><td>Unit</td><td>0.0001°</td></tr>
<tr><td>Setting range</td><td>–1200000 to 300000</td></tr>
<tr><td rowspan="4">RL75</td><td colspan="2">5th-axis position to permit robot access to the machine</td><td rowspan="4">Sets the 5th-axis position where the robot can enter inside the machine.
This position is checked as one of the conditions to turn the “DO 109: Robot access to machine permission” signal ON.

Default setting: 0</td></tr>
<tr><td>Conditions</td><td>Restarting of the machine</td></tr>
<tr><td>Unit</td><td>0.0001°</td></tr>
<tr><td>Setting range</td><td>0 to 3599999</td></tr>
<tr><td rowspan="4">TIM169</td><td colspan="2">Field network robot communication error check timer</td><td rowspan="4">Sets the time lapse to cause timeout when switching of the communication check signal was not detected.

Default setting: 100</td></tr>
<tr><td>Conditions</td><td>Reset</td></tr>
<tr><td>Unit</td><td>0.1 s</td></tr>
<tr><td>Setting range</td><td>0 or larger</td></tr>
<tr><td rowspan="4">F150</td><td colspan="2">NC basic information for field network
Output data offset</td><td rowspan="4">Sets the starting address of the initial output target register as a part of the NC basic information (R18176 to R18463) for the field network.

Example :

1. When F150 = 0


2. When F150 = 352


Default setting: 352 (Common to EtherNet/IP, PROFIBUS-DP, CC-Link)</td></tr>
<tr><td>Conditions</td><td>Restarting of the machine</td></tr>
<tr><td>Unit</td><td>Set at the NC</td></tr>
<tr><td>Setting range</td><td>0 or larger</td></tr>
</table>

8-5 Electrical Circuits (Interface)

- For details on the electrical circuits, refer to the electrical wiring diagrams for the robot interface.

8-6 Robot Online

This function is used to enable the robot interface.

Press and light the button on the machine operating panel to set the robot interface valid.

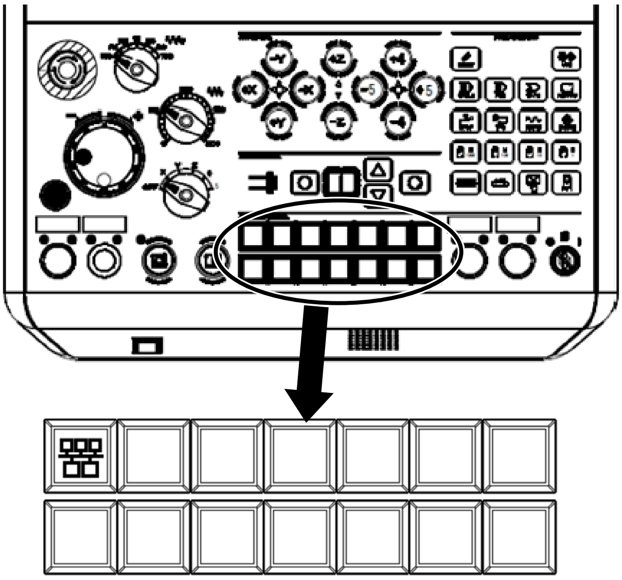


Fig. 8-4 Machine operating panel (MAZATROL SmoothX specifications)

8-7 Parameters Related to Field Network Input/Output Addresses (R Registers)

For field network specification models, the area in which input and output addresses (R registers) are used varies depending on the system configuration, so it needs to be specified with a parameter.

By factory default, the parameter is set on the assumption that the interface between the machine and the robot is assigned to station 1.

If either of the following conditions is met, the parameter needs to be changed on start-up of the system.

- If the interface between the machine and the robot is assigned to station 2 or later.
- In CC-Link, if the system contains the robot, the machine and other devices.

Change the parameter to the value obtained by adding the data size of each system to the default set value, referring to the following subsections.

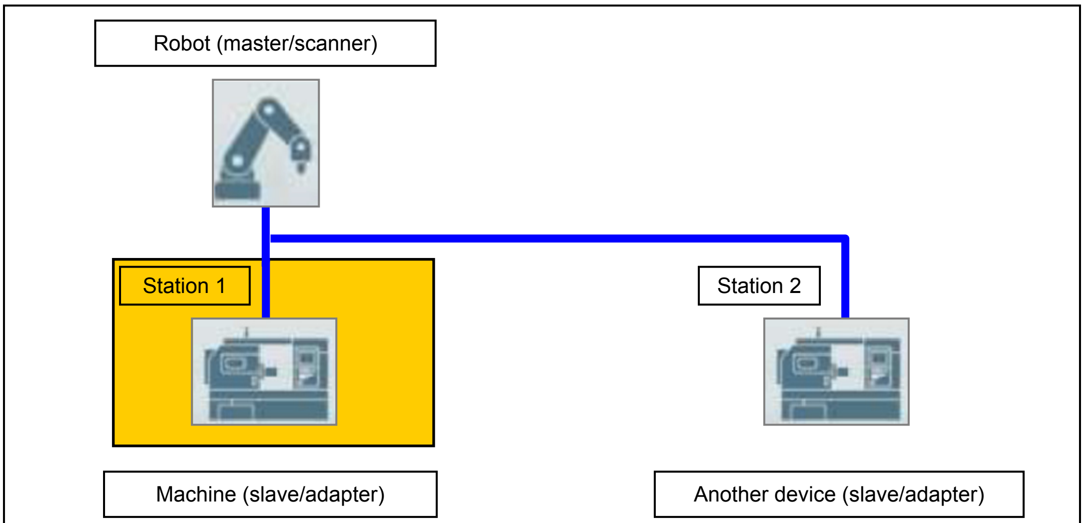
Note that the explanation below covers the R register when using CH1.

	• For details of parameters, refer to 8-4 “Parameters”.

8-7-1 For EtherNet/IP and PROFIBUS-DP

The setting of parameters **RS224** and **RS225** need to be changed according to the system configuration.

Example 1: System containing the machine and another device



Station 1: Interface between the robot and the machine

- Data size: 100 bytes (50 words)

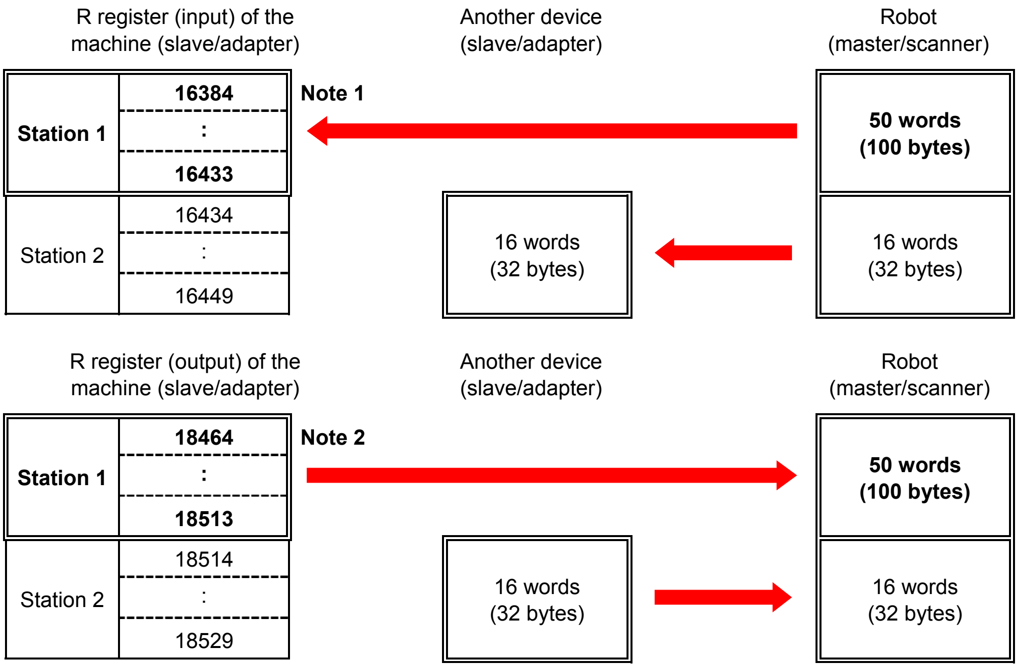
Station 2: Interface between the robot and another device

- Data size: 32 bytes (16 words)

Parameter		Set value	
RS224	Robot interface advance Initial address of input data	16386	To be changed
RS225	Robot interface advance Initial address of output data	18178	To be changed

- The parameter settings need to be changed from the initial value because R16384 to R16385 are occupied by the NC.

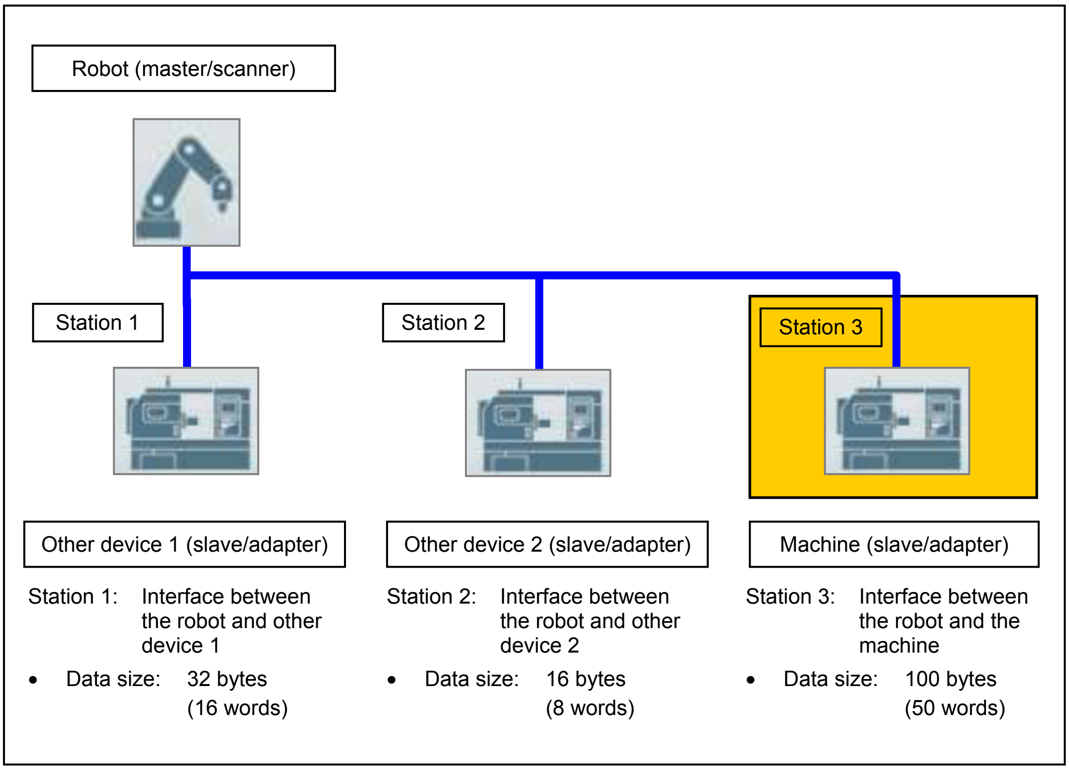
< Address map >



Note 1: Address to be set in the parameter **RS224**

Note 2: Address to be set in the parameter **RS225**

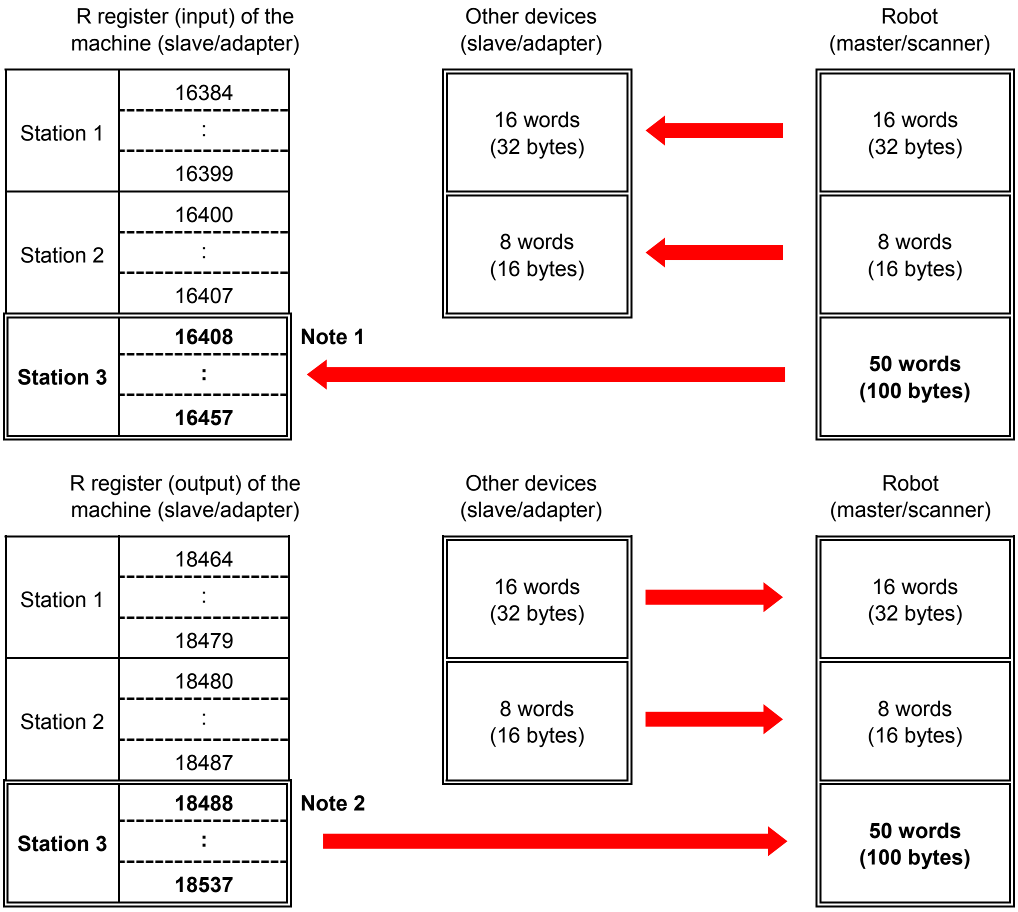
Example 2: System containing the machine and multiple other devices



- Total data size of stations 1 and 2: X = 16 words + 8 words = 24 words

Parameter		Set value	
RS224	Robot interface advance Initial address of input data	16410	16384 (Default set value) + X = 16408 + 2
RS225	Robot interface advance Initial address of output data	18202	18176 (Default set value) + X = 18200 + 2

< Address map >



Note 1: Address to be set in the parameter **RS224**

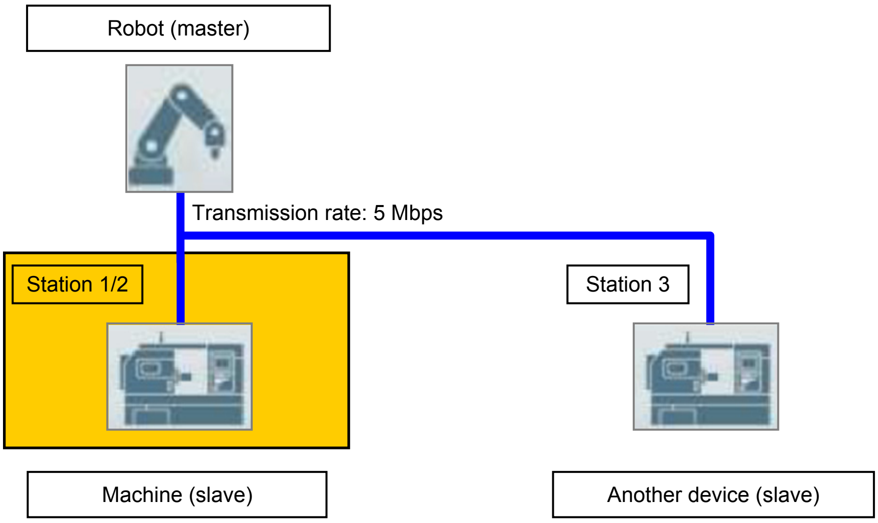
Note 2: Address to be set in the parameter **RS225**

8-7-2 For CC-Link

The parameters **CM1**, **CM2**, **CM13**, **CM14**, **RS224**, **RS225**, **RS226** and **RS227** need to be set depending on the system configuration.

- For the number of exclusive stations and the data size for each extended cyclic setting, refer to the separate manual Field Network Function (Option).

Example 1: System containing the machine and another device (with no remote I/O station)



Station 1/2: Interface between the robot and the machine

- Number of exclusive stations/extended cyclic setting: 2 stations/4 times
- Remote input/output data size: 192 points (12 words)
- Remote register data size: 32 words

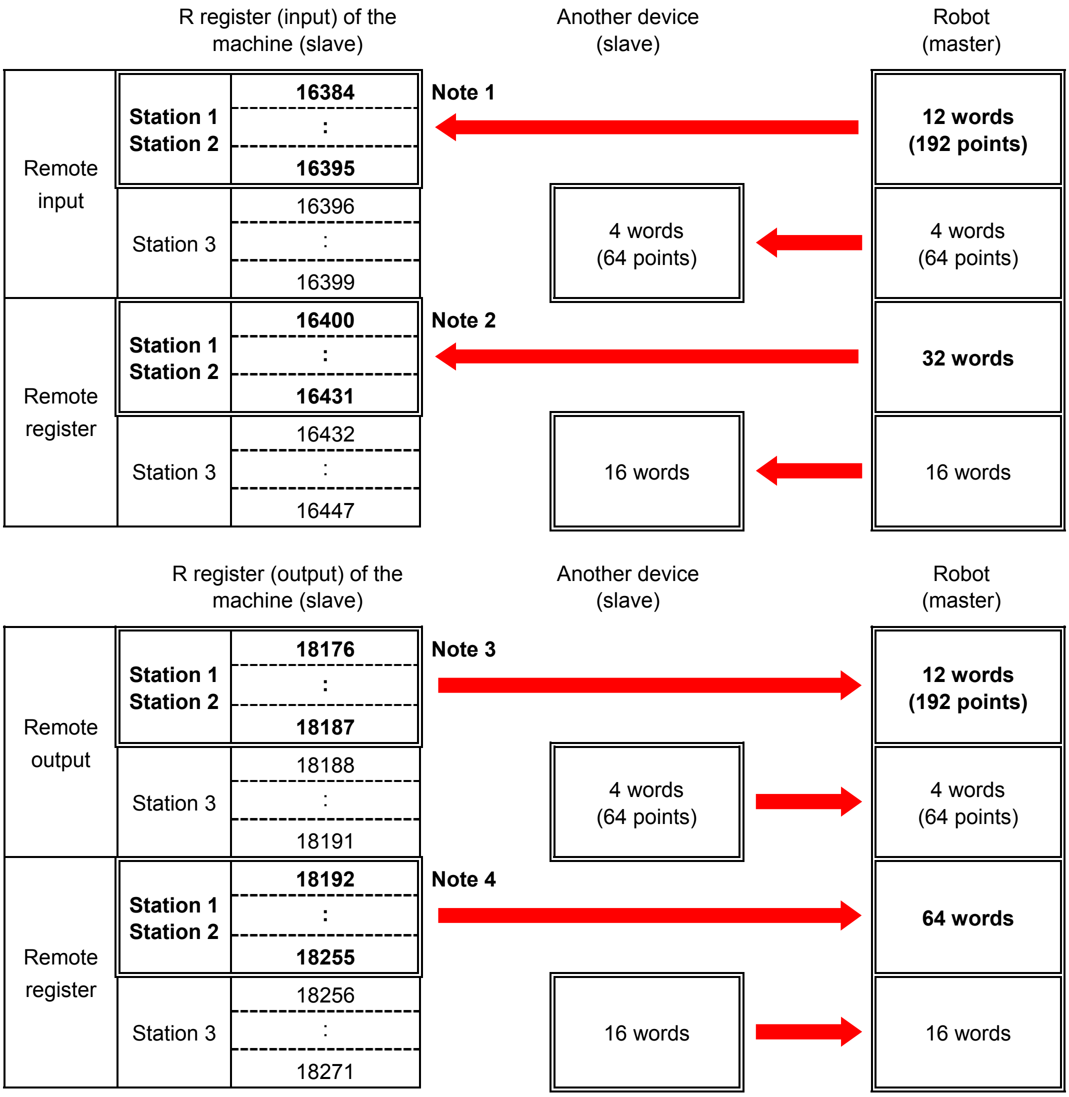
Station 3: Interface between the robot and another device

- Number of exclusive stations/extended cyclic setting: 1 station/4 times
- Remote input/output data size: 64 points (4 words)
- Remote register data size: 16 words

- Total remote input/output data size of station 3: X = 4 words

Parameter		Set value	
CM1	Station number	1	Station 1
CM2	Setting of transmission rate and mode	3	Transmission rate: 5 Mbps
CM13	Number of remote register (RWr) refresh register	16412	16408 (Default set value) + X = 16412
CM14	Number of remote register (RWw) refresh register	18204	18200 (Default set value) + X = 18204
RS224	Robot interface advance Initial address of input data	16384	Default set value
RS225	Robot interface advance Initial address of output data	18176	Default set value
RS226	Robot interface advance Initial address of input word data (for CC-Link)	16412	16408 (Default set value) + X = 16412
RS227	Robot interface advance Initial address of output word data (for CC-Link)	18204	18200 (Default set value) + X = 18204

< Address map >



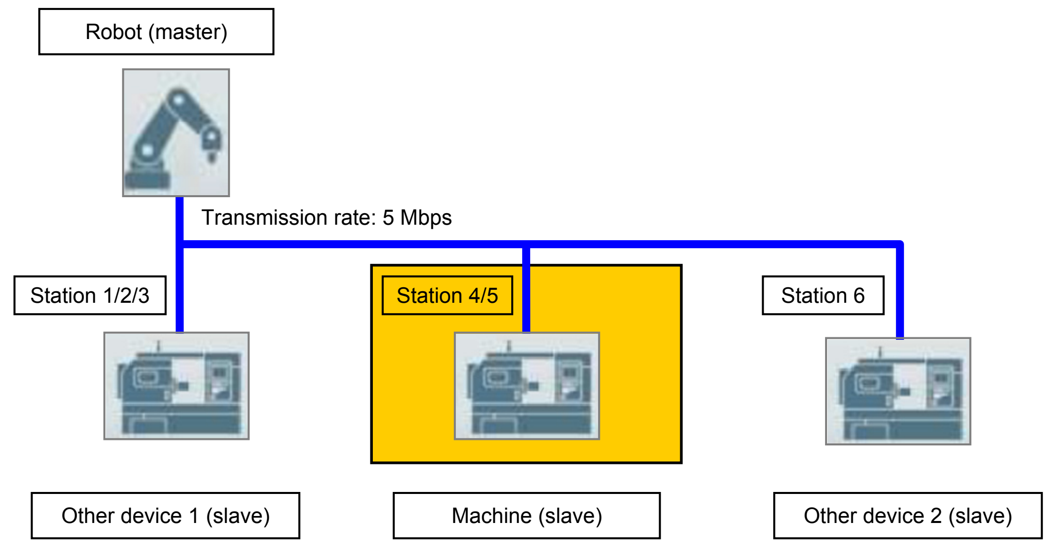
Note 1: Address to be set in the parameter **RS224**

Note 2: Address to be set in the parameter **CM13**/**RS226**

Note 3: Address to be set in the parameter **RS225**

Note 4: Address to be set in the parameter **CM14**/**RS227**

Example 2: System containing the machine and multiple other devices (with no remote I/O station)



Station 1/2/3: Interface between the robot and other device 1

- Number of exclusive stations/extended cyclic setting: 3 stations/2 times
- Remote input/output data size: 160 points (10 words)
- Remote register data size: 24 words

Station 4/5: Interface between the robot and the machine

- Number of exclusive stations/extended cyclic setting: 2 stations/4 times
- Remote input/output data size: 192 points (12 words)
- Remote register data size: 32 words

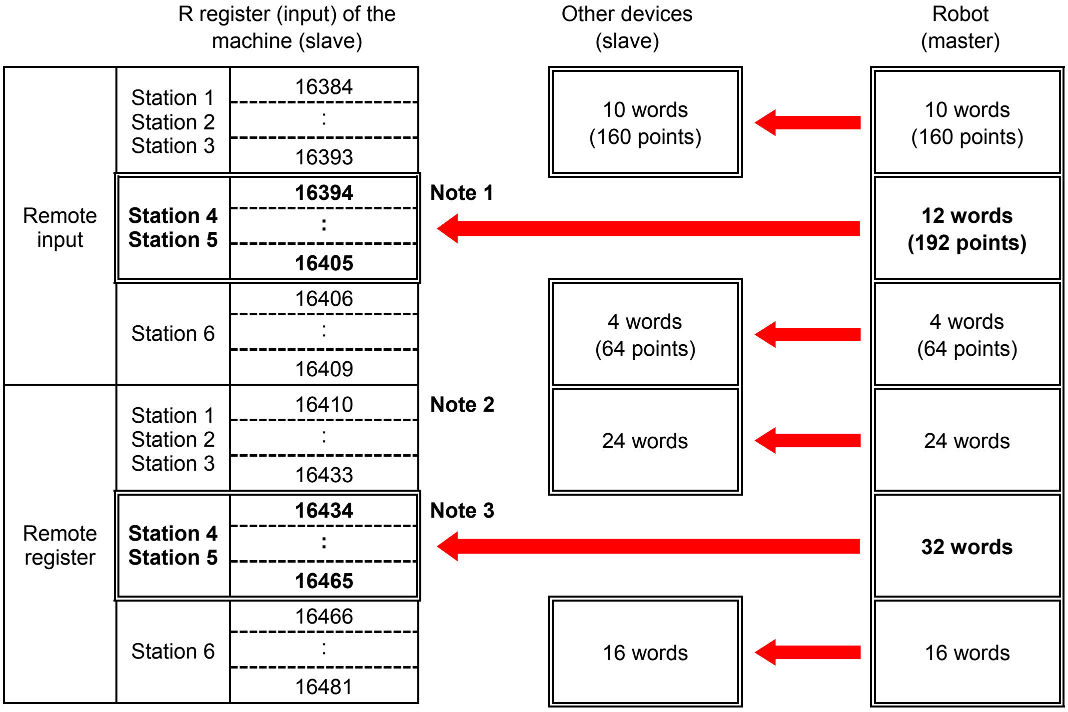
Station 6: Interface between the robot and other device 2

- Number of exclusive stations/extended cyclic setting: 1 station/4 times
- Remote input/output data size: 64 points (4 words)
- Remote register data size: 16 words

- Total size of remote input/output data of stations 1 through 3: X = 10 words
- Total size of remote input/output data of station 6: Y = 4 words
- Total size of remote register data of stations 1 through 3: Z = 24 words

Parameter		Set value	
CM1	Station number	4	Station 4
CM2	Setting of transmission rate and mode	3	Transmission rate: 5 Mbps
CM13	Number of remote register (RWr) refresh register	16422	16408 (Default set value) + X + Y = 16422
CM14	Number of remote register (RWw) refresh register	18214	18200 (Default set value) + X + Y = 18214
RS224	Robot interface advance Initial address of input data	16394	16384 (Default set value) + X = 16394
RS225	Robot interface advance Initial address of output data	18186	18176 (Default set value) + X = 18186
RS226	Robot interface advance Initial address of input word data (for CC-Link)	16446	16408 (Default set value) + X + Y + Z = 16446
RS227	Robot interface advance Initial address of output word data (for CC-Link)	18238	18200 (Default set value) + X + Y + Z = 18238

< Address map (input) >

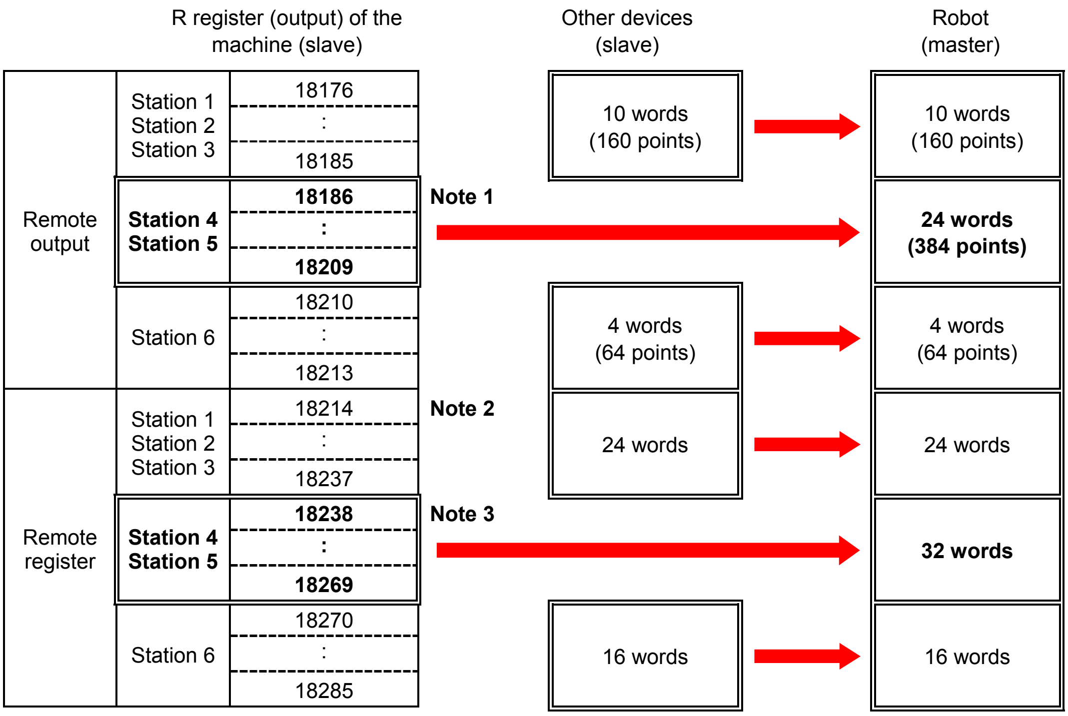


Note 1: Address to be set in the parameter **RS224**

Note 2: Address to be set in the parameter **CM13**

Note 3: Address to be set in the parameter **RS226**

< Address map (output) >

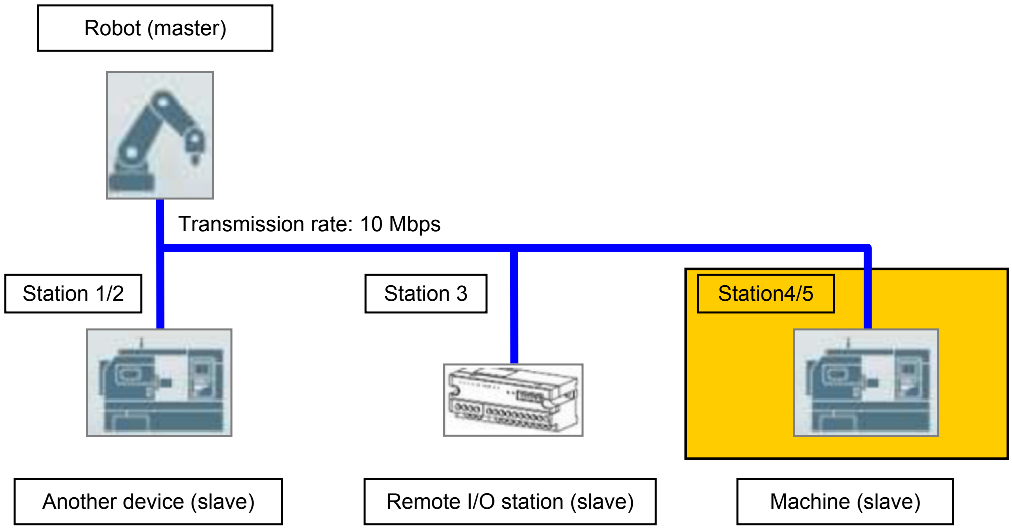


Note 1: Address to be set in the parameter **RS225**

Note 2: Address to be set in the parameter **CM14**

Note 3: Address to be set in the parameter **RS227**

Example 3: System containing the machine and multiple other devices (with remote I/O station)



Station 1/2: Interface between the robot and another device

- Number of exclusive stations/extended cyclic setting: 2 stations/2 times
- Remote input/output data size: 96 points (6 words)
- Remote register data size: 16 words

Station 3: Interface between the robot and remote I/O station

- Number of exclusive stations/extended cyclic setting: 1 stations/2 times
- Remote input/output data size: 32 points (2 words)
- Remote register data size: 0 words

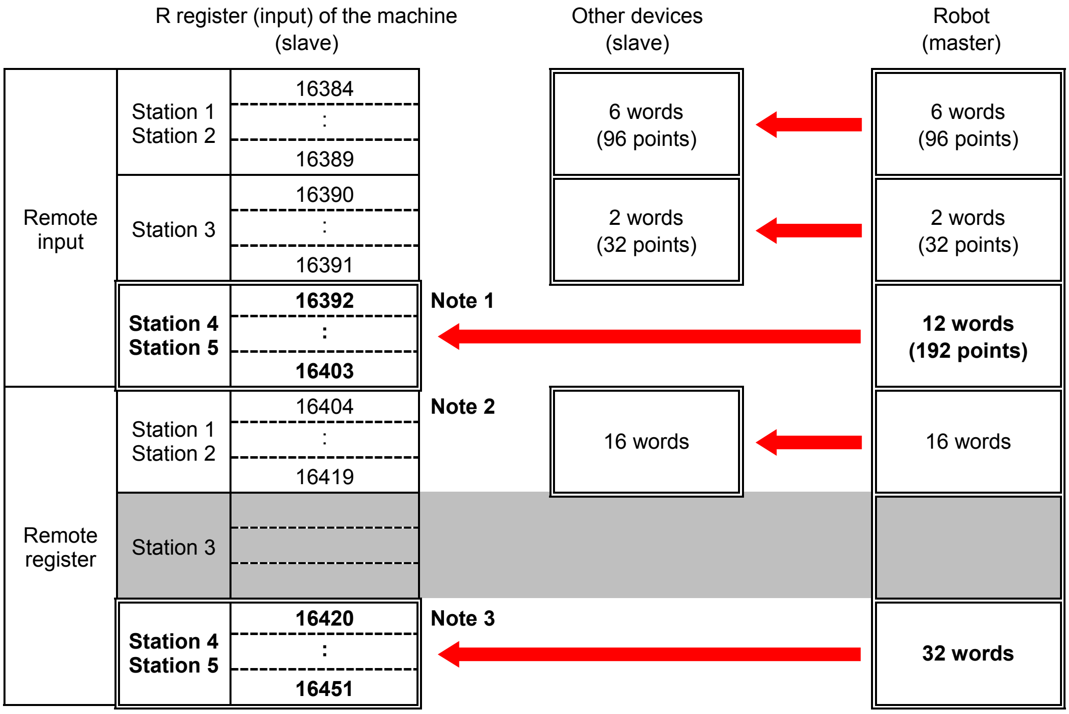
Station 4/5: Interface between the robot and the machine

- Number of exclusive stations/extended cyclic setting: 2 stations/4 times
- Remote input/output data size: 192 points (12 words)
- Remote register data size: 32 words

- Total size of remote input/output data of stations 1 through 3: X = 6 + 2 = 8 words
- Total size of remote register data of stations 1 through 3: Y = 16 + 0 = 16 words

Parameter		Set value	
CM1	Station number	4	Station 4
CM2	Setting of transmission rate and mode	4	Transmission rate: 10 Mbps
CM13	Number of remote register (RWr) refresh register	16416	16408 (Default set value) + X = 16416
CM14	Number of remote register (RWw) refresh register	18208	18200 (Default set value) + X = 18208
RS224	Robot interface advance Initial address of input data	16392	16384 (Default set value) + X = 16392
RS225	Robot interface advance Initial address of output data	18184	18176 (Default set value) + X = 18184
RS226	Robot interface advance Initial address of input word data (for CC-Link)	16432	16408 (Default set value) + X + Y = 16432
RS227	Robot interface advance Initial address of output word data (for CC-Link)	18224	18200 (Default set value) + X + Y = 18224

< Address map (input) >

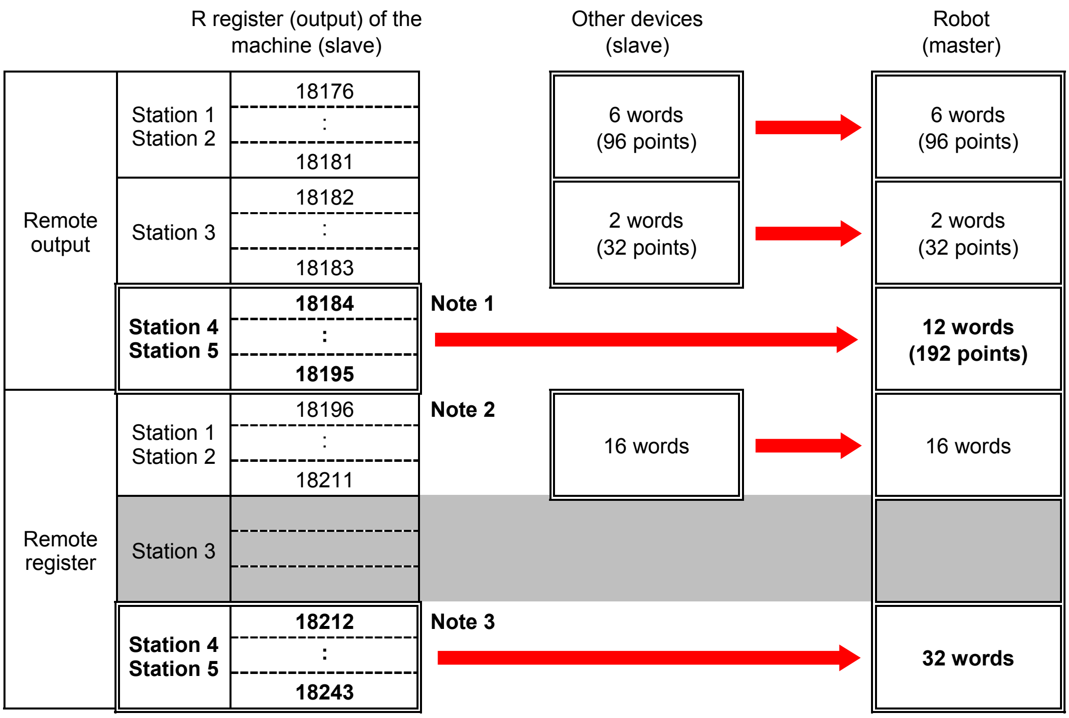


Note 1: Address to be set in the parameter **RS224**

Note 2: Address to be set in the parameter **CM13**

Note 3: Address to be set in the parameter **RS226**

< Address map (output) >

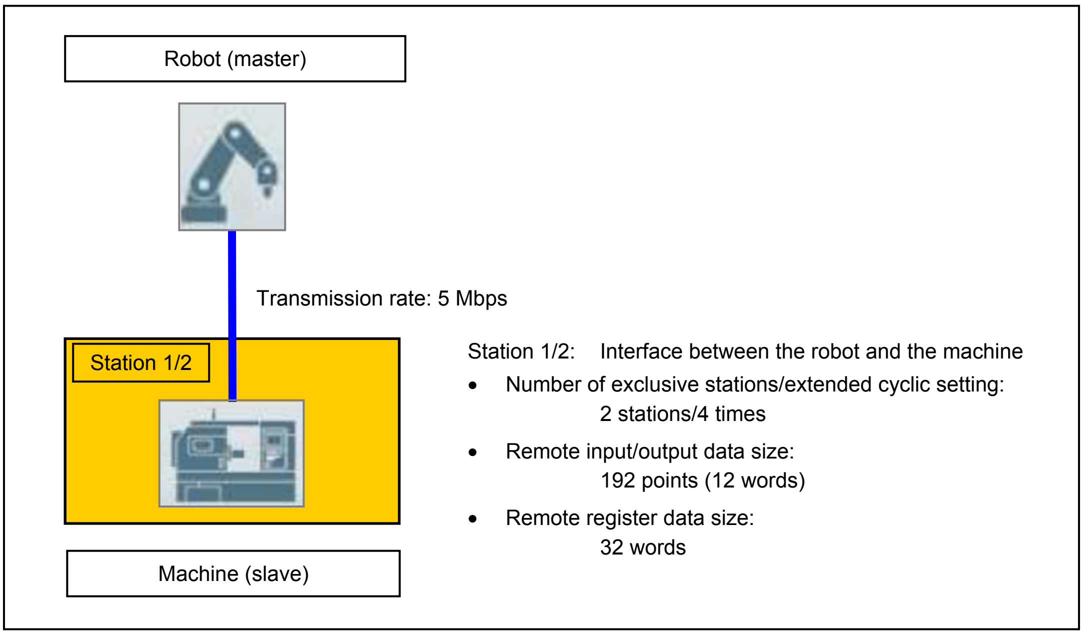


Note 1: Address to be set in the parameter **RS225**

Note 2: Address to be set in the parameter **CM14**

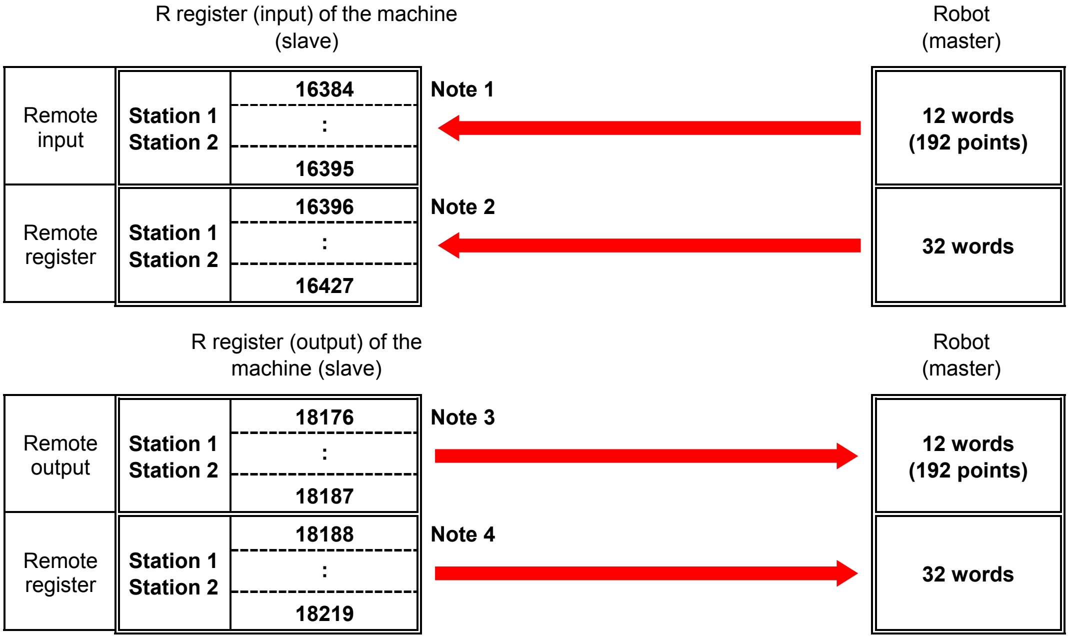
Note 3: Address to be set in the parameter **RS227**

Example 4: System containing the machine only



Parameter		Set value	
CM1	Station number	1	Station 1
CM2	Setting of transmission rate and mode	3	Transmission rate: 5 Mbps
CM13	Number of remote register (RWr) refresh register	16408	Default set value
CM14	Number of remote register (RWw) refresh register	18200	Default set value
RS224	Robot interface advance Initial address of input data	16384	Default set value
RS225	Robot interface advance Initial address of output data	18176	Default set value
RS226	Robot interface advance Initial address of input word data (for CC-Link)	16408	Default set value
RS227	Robot interface advance Initial address of output word data (for CC-Link)	18200	Default set value

< Address map >



Note 1: Address to be set in the parameter **RS224**

Note 2: Address to be set in the parameter **CM13**/**RS226**

Note 3: Address to be set in the parameter **RS225**

Note 4: Address to be set in the parameter **CM14**/**RS227**